Exact Finite-Sample Distribution and Moments of the Least Squares Estimator in First-Order Autoregressions: $O(1)$ Algorithms

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Abstract

We derive the exact finite-sample distribution function, density and moments (when they exist) of the least squares estimator of the coefficient in a Gaussian first-order autoregression. The intercept may be known or unknown and the initial observation fixed or stationary. With a fixed initial observation, the results cover every real coefficient, including the unit root and the explosive region. The joint characteristic function of the numerator and denominator of the estimation error follows from a bordered, rank-one-corrected tridiagonal Toeplitz determinant and a noncentral exponential factor. Five scalar primitives give this function, and all the derivatives that the inversion and moment formulas require, in closed form, at a cost independent of sample size. Chebyshev expansions resolve confluence near the unit root, accretivity settles the branch of the complex square root, and rearranged closed forms avoid cancellation in the explosive region. The same tools evaluate the local-to-unity limits of all four cases, including the Dickey–Fuller limits, and permit comparison with the closed-form explosive limit. In our benchmarks, exact p-values take milliseconds, and exact critical values and moments under a tenth of a second, at sample sizes from 10^2 to 10^6.

Keywords: autoregression; exact distribution; unit root; explosive process; ratio of quadratic forms; tridiagonal Toeplitz matrix; characteristic function inversion.
JEL classification: C22, C46, C63.

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1 Introduction

The least squares estimator $\hat{\phi}$ of the autoregressive coefficient ϕ in a first-order autoregression is one of the most studied objects in econometrics. Its bias was approximated and expanded by Hurwicz (1950), Kendall (1954), Marriott and Pope (1954), White (1961), Shenton and Johnson (1965), Phillips (1977), Bao and Ullah (2007), Bao (2007) and Kiviet and Phillips (2012), among many others. Its limiting distribution at and near the unit root was characterized by Rao (1978), Dickey and Fuller (1979), Evans and Savin (1981b), Chan and Wei (1987), Phillips (1987), Perron (1989), Nabeya and Tanaka (1990), Abadir (1993) and Larsson (1995), with a fixed nonzero or a stationary initial condition by Tanaka (1990), Perron (1991a,b) and Nabeya and Sørensen (1994), and in the explosive region by White (1958) and Anderson (1959). Its exact distribution underlies the median-unbiased estimators of Andrews (1993) and Andrews and Chen (1994); its finite-sample and local-to-unity distributions are the basis of the confidence intervals of Stock (1991) and Hansen (1999), the indirect-inference bias corrections of Phillips (2012) and the uniform inference procedures of Mikusheva (2007).

The estimation error $\hat{\phi}-\phi$ is a ratio of quadratic forms in Gaussian variables, so inversion formulas (Geary, 1944; Gil-Pelaez, 1951; Imhof, 1961) and moment representations (Sawa, 1978; Magnus, 1986; Bao and Kan, 2013) give its exact distribution and moments (see also Forchini, 2002; Ali, 2002). The exact distribution was expressed as an inversion integral by White (1958, 1959) for the explosive case and computed by Phillips (1977, 1978), by Evans and Savin (1981a, 1984) at the unit root, and by Tsui and Ali (1992, 1994) and Vinod and Shenton (1996) in the stationary case. Their evaluation meets four obstacles. The first is cost. Evaluating the characteristic function requires the determinant of an $n \times n$ matrix, where n is the number of observations in the regression. A dense determinant or eigendecomposition takes $O(n^3)$ operations, and the tridiagonal recursions of Evans and Savin (1984) and Tsui and Ali (1994) reduce this to $O(n)$, but the cost still grows with n at every point of the quadrature. The second is the branch of the complex square root, which Evans and Savin (1984) and Perron (1989) settled by a numerical search for discontinuities. The third is derivatives, which the density and the moments need and which the earlier literature derived by hand, case by case. The fourth, in the

explosive region, is cancellation. Partly as a result, the finite-sample tables in use — most prominently the Dickey–Fuller critical values in Fuller (1996) and the response surfaces of MacKinnon (1996) — rest on simulation.

Our first contribution is an evaluation of the joint characteristic function of the numerator and denominator, and of all the derivatives that the inversion and moment formulas need, in $O(1)$ operations per quadrature point at fixed derivative order and working precision. We cover four cases, combining a known or estimated intercept with a fixed or stationary initial observation. With a fixed start, every real ϕ is allowed. The matrix of the Gaussian integral has a tridiagonal Toeplitz core with a rank-one demeaning correction and one or two borders, and classical determinant and inverse formulas (Hu and O'Connell, 1996; da Fonseca and Petronilho, 2001), the Sherman–Morrison formula and a bordered-determinant lemma reduce it to five scalar primitives, so that the same assembly applies at $n = 20$ and at $n = 10^6$. The derivatives come from Taylor-mode arithmetic (Griewank and Walther, 2008), which carries every scalar of the assembly as a truncated power series and replaces the case-specific recursions. The moments then follow from Sawa's representation, centered at any point, which gives the central moments at the unit root without the catastrophic cancellation of raw moments. Kan et al. (2026) use this bordered Toeplitz framework for predictive regression; we extend it to the autoregressive cases, especially fixed starts, unit roots and explosive coefficients.

Our second contribution is to make the closed forms numerically reliable. Near the unit root the characteristic roots coalesce, and Chebyshev expansions evaluate the primitives without subtracting nearly equal roots. Along an inversion path the determinant circles the origin many times, so its principal square root changes sign at every crossing of the negative real axis. We prove that the matrices are accretive (complex symmetric with positive definite real part) on each case's parameter domain, including unit roots and explosive coefficients, and assemble the phase from factors that avoid the branch cut. In the explosive region a determinant of order ϕ^{2n} meets a correction of order ϕ^{-2n}, and direct subtraction loses about $2n \log_{10} |\phi|$ digits. Rearranged closed forms remove this exponential cancellation, and, written in scaled arguments, they give the distribution of the normalized estimator for every sample size.

Our third contribution concerns the limiting distributions, which the same tools eval-

uate. The local-to-unity limits of all four cases, including the Dickey–Fuller distributions, have characteristic functions known in closed form (Phillips, 1987; Tanaka, 1990; Perron, 1991a,b; Nabeya and Sørensen, 1994; Haldrup and Hylleberg, 1995), and we settle their branch by the same device. The limit in the explosive case is of Cauchy type (White, 1958; Anderson, 1959; Wang and Yu, 2015), and with an intercept we give an explicit finite-sample correction to it.

The rest of the paper is organized as follows. Section 2 sets out the model, the four cases and the inversion formulas; Section 3 derives the joint characteristic functions and their $O(1)$ evaluation; Section 4 treats the numerical issues; Section 5 the exact moments; Section 6 the limiting distributions; Section 7 reports numerical results, and Section 8 concludes. The main proofs are in Appendix A; the remaining proofs, additional derivations and additional tables are in the Online Supplement, and the MATLAB programs are described in the user manual distributed with them.

Throughout, $\mathbf{1}_m$ and $\mathbf{0}_m$ are the $m \times 1$ vectors of ones and of zeros, $\boldsymbol{I}_m$ the identity matrix of order m, $\boldsymbol{e}_{j,m}$ the j-th column of $\boldsymbol{I}_m$, and $\boldsymbol{M}_m = \boldsymbol{I}_m - \mathbf{1}_m\mathbf{1}_m'/m$ the demeaning matrix. We write $\mathrm{i} = \sqrt{-1}$. For a complex number z, Re z and Im z are its real and imaginary parts, $\mathrm{Arg}(z) \in (-\pi, \pi]$ is its principal argument and $z^{1/2}$ its principal square root; for a complex matrix $\boldsymbol{K}$, Re $\boldsymbol{K}$ and Im $\boldsymbol{K}$ are its entrywise real and imaginary parts, and a complex symmetric $\boldsymbol{K}$ with Re $\boldsymbol{K}$ positive definite is called *accretive*. A prime denotes the transpose without complex conjugation, for complex as well as real vectors and matrices, and $\boldsymbol{z}^*$ denotes the conjugate transpose.

We use $\mathbb{E}$ and $\mathbb{P}$ for expectation and probability, Var, Cov, Corr and sd for variance, covariance, correlation and standard deviation, and $\mathcal{N}(\mu, v)$ for a normal distribution with mean μ and variance v. The standard normal distribution function is denoted by $\Phi_{\mathcal{N}}$. For a square matrix $\boldsymbol{A}$, $|\boldsymbol{A}| = \det \boldsymbol{A}$; for a scalar z, $|z|$ is its modulus. Thus $|\det \boldsymbol{A}|$ is a determinant's modulus. We write $\|\boldsymbol{z}\| = (\boldsymbol{z}^*\boldsymbol{z})^{1/2}$, $\bar{z}$ for complex conjugation, $\mathbf{1}\{\cdot\}$ for an indicator, $\lfloor x \rfloor$ for the greatest integer not exceeding x, and $\mathrm{sgn}(x)$ for the sign of a real number. The symbols $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{Z}$ denote the real numbers, complex numbers and integers. The symbols $\Rightarrow$ and $\overset{d}{=}$ denote convergence in distribution and equality in distribution, respectively, $\perp$ means independence, and O_p means bounded order in probability. The deterministic symbols O and o have their usual order meanings in the

stated limit. Finally, $\chi^2_\nu(\delta)$ denotes a noncentral chi-square distribution with ν degrees of freedom and noncentrality δ, and χ^2_ν the central one.

2 The Model, Four Cases, and Inversion Formulas

We specify the Gaussian autoregression and the two least squares estimators, then combine the intercept specification with the initial condition to define four cases. Their estimation errors share a ratio representation, which yields common inversion and moment formulas.

2.1 The model and the estimators

Consider the first-order autoregression

$$y_t = \alpha + \phi y_{t-1} + \sigma e_t, \qquad t = 1, \ldots, n, \tag{1}$$

where y_t is the observation at time t, $\alpha \in \mathbb{R}$ is the intercept, $\phi \in \mathbb{R}$ is the autoregressive coefficient, $\sigma > 0$ is the innovation standard deviation, and $n \geq 3$ is the number of regression observations; $e_1, \ldots, e_n$ are independent standard normal random variables and y_0 is either a fixed constant or, when $|\phi| < 1$, a draw from the stationary distribution $\mathcal{N}(\alpha/(1-\phi), \sigma^2/(1-\phi^2))$ independent of the e_t. Two least squares estimators of ϕ are in common use. When the intercept is zero and this is imposed, the case we label "known" below, ϕ is estimated by the regression of y_t on y_{t-1} through the origin,

$$\hat{\phi} = \frac{\sum_{t=1}^{n} y_{t-1} y_t}{\sum_{t=1}^{n} y_{t-1}^2}; \tag{2}$$

when it is unknown, by the regression of y_t on a constant and y_{t-1},

$$\hat{\phi} = \frac{\sum_{t=1}^{n} (y_{t-1} - \bar{y}_-) y_t}{\sum_{t=1}^{n} (y_{t-1} - \bar{y}_-)^2}, \qquad \bar{y}_- = \frac{1}{n} \sum_{t=1}^{n} y_{t-1}. \tag{3}$$

For a stationary process the known case is that of a known mean, with the data centered at that mean. We use the same symbol for both estimators; which one is meant is fixed

by the case, defined below. For both estimators the estimation error is a ratio,

$$\hat{\phi} - \phi = \frac{u}{v}, \tag{4}$$

with, for (2), $u = \sigma \sum_{t=1}^{n} y_{t-1} e_t$ and $v = \sum_{t=1}^{n} y_{t-1}^2$, and, for (3), $u = \sigma \sum_{t=1}^{n} (y_{t-1} - \bar{y}_-) e_t$ and $v = \sum_{t=1}^{n} (y_{t-1} - \bar{y}_-)^2$, where we have used $\sum_{t=1}^{n} (y_{t-1} - \bar{y}_-) = 0$ to remove the intercept from the numerator. In every case u and v are quadratic functions of a Gaussian vector, with $v > 0$ almost surely.

The four cases of the paper are the combinations of the two estimators with the two initial conditions. We label them by two letters, the first for the initial observation (Deterministic, that is, a fixed value, or drawn from its Stationary distribution) and the second for the intercept (Known or Unknown), and record the parameters on which the distribution of $\hat{\phi}$ depends:

DK. y_0 fixed, $\alpha = 0$, estimator (2). Dividing the data by σ leaves $\hat{\phi}$ unchanged, so its distribution depends on (n, ϕ, z_0) with $z_0 = y_0/\sigma$. Any real ϕ is allowed.

DU. y_0 fixed, α unknown, estimator (3). The estimator is invariant to a common shift and a common scaling of the data, so with $w_t = (y_t - y_0)/\sigma$ the model becomes $w_t = \gamma + \phi w_{t-1} + e_t$, $w_0 = 0$, and the distribution of $\hat{\phi}$ depends on (n, ϕ, γ) with

$$\gamma = \frac{\alpha - (1 - \phi) y_0}{\sigma}, \tag{5}$$

the parameterization of Evans and Savin (1984). Any real ϕ is allowed; when $\phi = 1$, $\gamma = \alpha/\sigma$ is the standardized drift of the random walk.

SK. $y_0 \sim \mathcal{N}(0, \sigma^2/(1 - \phi^2))$, $\alpha = 0$, estimator (2); $|\phi| < 1$; the distribution depends on (n, ϕ).

SU. $y_0 \sim \mathcal{N}(\alpha/(1-\phi), \sigma^2/(1-\phi^2))$, α unknown, estimator (3); $|\phi| < 1$; the distribution depends on (n, ϕ).

Case DK is the setting of White (1958, 1959), Anderson (1959), Phillips (1977) and Evans and Savin (1981a); case DU that of Evans and Savin (1984), Nankervis and Savin

(1985) and Andrews (1993); cases SK and SU those of Tsui and Ali (1992), Andrews (1993), Tsui and Ali (1994) and Vinod and Shenton (1996).

2.2 Inversion and moment formulas

For real transform arguments $(t_1, t_2) \in \mathbb{R}^2$, let $\varphi(t_1, t_2) = \mathbb{E}[\exp(\mathrm{i}t_1 u + \mathrm{i}t_2 v)]$ be the joint characteristic function of the numerator and denominator in (4). For a real estimation-error threshold r, since $v > 0$ almost surely, $\mathbb{P}[\hat{\phi} - \phi < r] = \mathbb{P}[u - rv < 0]$, and the inversion theorem of Gil-Pelaez (1951) in the form used by Imhof (1961) gives the cumulative distribution function (cdf)

$$F(r) \equiv \mathbb{P}[\hat{\phi} - \phi < r] = \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \frac{\operatorname{Im} \varphi(t_1, -rt_1)}{t_1} \, \mathrm{d}t_1. \tag{6}$$

Differentiating under the integral sign (Geary, 1944; Broda and Kan, 2016) gives the probability density function (pdf),

$$f(r) = \frac{1}{\pi} \int_0^\infty \operatorname{Im} \left(\frac{\partial \varphi(t_1, t_2)}{\partial t_2} \bigg|_{t_2 = -rt_1} \right) \mathrm{d}t_1. \tag{7}$$

The differentiation is legitimate when the integral in (7) converges absolutely, which is the case when the quadratic part of $u - rv$ has rank at least three, or when the linear term of $u - rv$, in centered and whitened Gaussian coordinates, has a nonzero projection onto the null space of that quadratic part (Section 4.4). Here the rank of a quadratic form means the rank of its coefficient matrix in nondegenerate Gaussian coordinates. In the admitted domain $n \geq 3$ both conditions fail only at $n = 3$: in case DK with $z_0 = 0$ at $\hat{\phi} = 0$, and in cases DU and SU at $\hat{\phi} = -1/2$, for all values of the parameters (Section S6 of the Online Supplement). There the density has a logarithmic singularity, and the programs return $+\infty$. For a specified probability $0 < p < 1$, the corresponding quantile is the solution of $F(r) = p$. For the moments, let $M(s_1, s_2) = \mathbb{E}[\exp(s_1 u + s_2 v)] = \varphi(-\mathrm{i}s_1, -\mathrm{i}s_2)$ be the joint moment generating function, which exists in a neighborhood of the origin and for every $s_2 \leq 0$ when s_1 is near zero. For a real centering constant c write $\hat{\phi} - c = (u - \tilde{c}v)/v$ with $\tilde{c} = c - \phi$, and let $M_c(s_1, s_2) = \mathbb{E}[e^{s_1(u - \tilde{c}v) + s_2 v}] = M(s_1, s_2 - s_1\tilde{c})$ be the joint moment generating function of $(u - \tilde{c}v, v)$. Since $v > 0$ with probability one and $M_c(s_1, 0)$ is finite

for s_1 near zero, Lemma 1 of Meng (2005) gives the representation of Sawa (1978) (see also Magnus, 1986) for every positive integer k for which the moment exists (Proposition 1 below):

$$\mathbb{E}[(\hat{\phi} - c)^k] = \frac{1}{(k-1)!} \int_0^\infty s_2^{k-1} \left. \frac{\partial^k M_c(s_1, -s_2)}{\partial s_1^k} \right|_{s_1=0} \mathrm{d}s_2. \tag{8}$$

The choice $c = 0$ gives the raw moments, $c = \phi$ the moments of the estimation error, and $c = \mathbb{E}[\hat{\phi}]$ the central moments.

Proposition 1 (Existence of moments)**.** *For real $k > 0$, $\mathbb{E}|\hat{\phi}|^k < \infty$ if and only if $k < n-1$ in cases* DU *and* SU *and in case* DK *with $z_0 = 0$; if and only if $k < n$ in case* SK*; and for every k in case* DK *with $z_0 \neq 0$.*

Where v can come arbitrarily close to zero, the condition is that k be smaller than the rank of v, which is n in case SK and $n-1$ in the other cases, where a fixed initial value or demeaning removes one dimension; in case DK with $z_0 \neq 0$ the denominator is bounded below by z_0^2 and the numerator has moments of all orders, so that all positive moments of $|\hat{\phi}|$ exist.

Formulas (6)–(8) reduce the whole problem to one object: the joint characteristic function $\varphi(t_1, t_2)$, together with its first derivative in t_2 and, for the moments, its derivatives of any order along the affine paths in (t_1, t_2) on which $M_c(s_1, -s_2)$ is evaluated. The next section shows how to evaluate all of these at a cost independent of n.

3 The Joint Characteristic Function and $O(1)$ Evaluation

The Gaussian integration formula expresses each joint characteristic function through a bordered Toeplitz determinant and a noncentral bilinear form. We reduce these objects to five scalar primitives and evaluate the same assembly in Taylor-mode arithmetic to obtain the derivatives needed for densities and moments.

3.1 Four determinants

Throughout this section $(t_1, t_2) \in \mathbb{R}^2$; Section 3.3 evaluates the same formulas, by analytic continuation, at the complex arguments of the moment generating function. We define the scalars

$$p = 1 + \phi^2 + 2\mathrm{i}(\phi t_1 - t_2), \qquad q = -(\phi + \mathrm{i}t_1), \tag{9}$$

and, for $k \geq 1$, the $k \times k$ symmetric tridiagonal Toeplitz matrix $\boldsymbol{D}_k$ with diagonal entries p and off-diagonal entries q; we write $\boldsymbol{D} = \boldsymbol{D}_{n-1}$ and set $|\boldsymbol{D}_0| = 1$. The same p, q and $\boldsymbol{D}$ appear in all four cases; the cases differ only in how $\boldsymbol{D}$ is bordered and corrected. Since the distribution of $\hat{\phi} - \phi$ depends on σ only through z_0 in case DK and γ in case DU (Section 2.1), we set $\sigma = 1$ without loss of generality.

Theorem 1 (Joint characteristic function)**.** *Let $m = n - 1$. The joint characteristic function of (u, v) in (4) is*

$$\varphi(t_1, t_2) = \varrho\, |\boldsymbol{K}|^{-1/2} \exp\left(\tfrac{1}{2}\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta} + \eta\right),$$

where the square root is the branch continuous in (t_1, t_2) with $\varphi(0, 0) = 1$, and where $\boldsymbol{K}$, $\boldsymbol{\beta}$, η and ϱ are as follows.

(i) Case DK*: $\varrho = 1$,*

$$\boldsymbol{K} = \begin{bmatrix} \boldsymbol{D} & q\boldsymbol{e}_{m,m} \\ q\boldsymbol{e}'_{m,m} & 1 \end{bmatrix}, \qquad \boldsymbol{\beta} = -qz_0 \begin{bmatrix} \boldsymbol{e}_{1,m} \\ 0 \end{bmatrix}, \qquad \eta = -\tfrac{1}{2}z_0^2(p-1). \tag{10}$$

(ii) Case DU*: $\varrho = 1$,*

$$\boldsymbol{K} = \begin{bmatrix} \boldsymbol{D} + \kappa \mathbf{1}_m \mathbf{1}'_m & \boldsymbol{b} \\ \boldsymbol{b}' & 1 \end{bmatrix}, \qquad \boldsymbol{b} = \frac{\mathrm{i}t_1}{n}\mathbf{1}_m + q\boldsymbol{e}_{m,m}, \qquad \kappa = \frac{2\mathrm{i}[(1-\phi)t_1 + t_2]}{n}, \tag{11}$$

and $\boldsymbol{\beta} = \gamma\,((1-\phi)\mathbf{1}'_m,\, 1)'$, $\eta = -\frac{n}{2}\gamma^2$.

(iii) Case SK*:* $\varrho = (1-\phi^2)^{1/2}$, $\boldsymbol{\beta} = \mathbf{0}_{n+1}$, $\eta = 0$,

$$\boldsymbol{K} = \begin{bmatrix} p-\phi^2 & q\boldsymbol{e}'_{1,m} & 0 \\ q\boldsymbol{e}_{1,m} & \boldsymbol{D} & q\boldsymbol{e}_{m,m} \\ 0 & q\boldsymbol{e}'_{m,m} & 1 \end{bmatrix}. \tag{12}$$

(iv) Case SU*:* $\varrho = (1-\phi^2)^{1/2}$, $\boldsymbol{\beta} = \mathbf{0}_{n+1}$, $\eta = 0$,

$$\boldsymbol{K} = \begin{bmatrix} 1 + 2\mathrm{i}(\phi t_1 - t_2)\big(1 - \frac{1}{n}\big) & \boldsymbol{a}' & \mathrm{i}t_1/n \\ \boldsymbol{a} & \boldsymbol{D} + \kappa \mathbf{1}_m \mathbf{1}'_m & \boldsymbol{b} \\ \mathrm{i}t_1/n & \boldsymbol{b}' & 1 \end{bmatrix}, \quad \kappa = \frac{2\mathrm{i}[(1-\phi)t_1 + t_2]}{n}, \tag{13}$$

where

$$\boldsymbol{a} = \frac{\mathrm{i}[(1-2\phi)t_1 + 2t_2]}{n}\mathbf{1}_m + q\boldsymbol{e}_{1,m}, \tag{14}$$

and $\boldsymbol{b}$ *is as in (11).*

In every case Re $\boldsymbol{K}$ *does not depend on* (t_1, t_2) *and is positive definite, for every real* ϕ *in cases* DK *and* DU *and for* $|\phi| < 1$ *in cases* SK *and* SU*:* Re $\boldsymbol{K} = (\boldsymbol{I}_n - \phi \boldsymbol{J}_n)'(\boldsymbol{I}_n - \phi \boldsymbol{J}_n)$ *in cases* DK *and* DU*, where* $\boldsymbol{J}_n$ *is the* $n \times n$ *shift matrix with ones on the first subdiagonal, and* Re $\boldsymbol{K} = (1-\phi^2)\boldsymbol{V}_{n+1}^{-1}$ *in cases* SK *and* SU*, where* $\boldsymbol{V}_{n+1}$ *is the* $(n+1)\times(n+1)$ *Toeplitz matrix with entries* $\phi^{|i-j|}$.

The proof is in Section S6 of the Online Supplement. The joint characteristic function is that of White (1958) in case DK, of Evans and Savin (1984) in case DU, where (11) is a compact form of their Eq. (3.9), and of Tsui and Ali (1994) in cases SK and SU, in different notation. What the theorem adds is a common form: in all four cases $\boldsymbol{K}$ is the same tridiagonal Toeplitz core $\boldsymbol{D}$ with one or two borders and, in cases DU and SU, a rank-one demeaning correction, and its real part is positive definite and does not depend on (t_1, t_2). The common assembly and the positive, transform-independent real part of $\boldsymbol{K}$ are what the subsequent reduction (Section 3.2) and branch analysis (Section 4.2) exploit.

3.2 Reduction to five primitives

All four matrices in Theorem 1 have the form

$$\boldsymbol{K} = \begin{bmatrix} c_{11} & \boldsymbol{a}' & c_{1n} \\ \boldsymbol{a} & \boldsymbol{\Delta} & \boldsymbol{b} \\ c_{1n} & \boldsymbol{b}' & 1 \end{bmatrix}, \quad \boldsymbol{\Delta} = \boldsymbol{D} + \kappa \mathbf{1}_m \mathbf{1}_m', \quad \boldsymbol{a} = a_1 \mathbf{1}_m + a_2 \boldsymbol{e}_{1,m}, \quad \boldsymbol{b} = b_1 \mathbf{1}_m + q \boldsymbol{e}_{m,m}, \tag{15}$$

with scalar coefficients $c_{11}, c_{1n}, a_1, a_2, b_1, \kappa$ listed in Table 1, which also gives $\boldsymbol{\beta}_c$ and β_n in the partition $\boldsymbol{\beta} = (0, \boldsymbol{\beta}_c', \beta_n)'$, where $\boldsymbol{\beta}_c$ is the m-dimensional core vector and β_n is the last-coordinate scalar. In cases DK and DU this representation appends a dummy first coordinate with diagonal entry one and zero couplings, which leaves both the determinant and the bilinear form unchanged; $\boldsymbol{K}$ and $\boldsymbol{\beta}$ then denote the augmented matrix and vector. Two lemmas reduce the determinant of (15) and the bilinear form $\boldsymbol{\beta}' \boldsymbol{K}^{-1} \boldsymbol{\beta}$ to the determinant of $\boldsymbol{D}$ and four bilinear forms of $\boldsymbol{D}^{-1}$.

Table 1: Coefficients of the bordered matrix (15) in the four cases

Case	c_{11}	c_{1n}	a_1	a_2	b_1	κ	$\boldsymbol{\beta}_c$	β_n
DK	1	0	0	0	0	0	$-qz_0 \boldsymbol{e}_{1,m}$	0
DU	1	0	0	0	$\frac{\mathrm{i}t_1}{n}$	$\frac{2\mathrm{i}[(1-\phi)t_1+t_2]}{n}$	$\gamma(1-\phi)\mathbf{1}_m$	γ
SK	$1+2\mathrm{i}(\phi t_1 - t_2)$	0	0	q	0	0	$\mathbf{0}_m$	0
SU	$1+2\mathrm{i}(\phi t_1 - t_2)(1-\frac{1}{n})$	$\frac{\mathrm{i}t_1}{n}$	$\frac{\mathrm{i}[(1-2\phi)t_1+2t_2]}{n}$	q	$\frac{\mathrm{i}t_1}{n}$	$\frac{2\mathrm{i}[(1-\phi)t_1+t_2]}{n}$	$\mathbf{0}_m$	0

Notes: In every case $p = 1 + \phi^2 + 2\mathrm{i}(\phi t_1 - t_2)$ and $q = -(\phi + \mathrm{i}t_1)$ are the diagonal and off-diagonal entries of the $(n-1) \times (n-1)$ core $\boldsymbol{D}$. The last two columns give the partition $\boldsymbol{\beta} = (0, \boldsymbol{\beta}_c', \beta_n)'$ of the linear term used in Lemma 1.

Lemma 1 (Bordered matrices)**.** *Let $\boldsymbol{K}$ be as in (15) with $\boldsymbol{\Delta}$ nonsingular, and define*

$$\delta_1 = c_{11} - \boldsymbol{a}' \boldsymbol{\Delta}^{-1} \boldsymbol{a}, \qquad \delta_2 = c_{1n} - \boldsymbol{a}' \boldsymbol{\Delta}^{-1} \boldsymbol{b}, \qquad \delta_3 = 1 - \boldsymbol{b}' \boldsymbol{\Delta}^{-1} \boldsymbol{b},$$
$$\boldsymbol{\Sigma} = \begin{bmatrix} \delta_1 & \delta_2 \\ \delta_2 & \delta_3 \end{bmatrix}, \qquad \delta_4 = |\boldsymbol{\Sigma}| = \delta_1 \delta_3 - \delta_2^2.$$

Then $|\boldsymbol{K}| = |\boldsymbol{\Delta}|\,\delta_4$*, and if* $\delta_4 \neq 0$ *(that is,* $\boldsymbol{K}$ *is nonsingular, as it is for all real* (t_1, t_2) *by Theorem 2), for a vector* $\boldsymbol{\beta} = (0, \boldsymbol{\beta}_c', \beta_n)'$ *partitioned conformably with* $\boldsymbol{K}$*,*

$$\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta} = \boldsymbol{\beta}_c'\boldsymbol{\Delta}^{-1}\boldsymbol{\beta}_c + \frac{\delta_3 g_a^2 + 2\delta_2 g_a(\beta_n - g_b) + \delta_1(\beta_n - g_b)^2}{\delta_4}, \tag{16}$$

where $g_a = \boldsymbol{a}'\boldsymbol{\Delta}^{-1}\boldsymbol{\beta}_c$ *and* $g_b = \boldsymbol{b}'\boldsymbol{\Delta}^{-1}\boldsymbol{\beta}_c$*.*

Lemma 2 (Tridiagonal Toeplitz primitives)**.** *Let* $\boldsymbol{D}$ *be the* $m \times m$ *(*$m \geq 2$*) nonsingular symmetric tridiagonal Toeplitz matrix with diagonal* p *and off-diagonal* q*, and let*

$$d_1 = \mathbf{1}_m'\boldsymbol{D}^{-1}\mathbf{1}_m, \quad d_2 = \boldsymbol{e}_{1,m}'\boldsymbol{D}^{-1}\mathbf{1}_m, \quad d_3 = \boldsymbol{e}_{1,m}'\boldsymbol{D}^{-1}\boldsymbol{e}_{1,m}, \quad d_4 = \boldsymbol{e}_{1,m}'\boldsymbol{D}^{-1}\boldsymbol{e}_{m,m}. \tag{17}$$

If $q = 0$ *(which happens at* $\phi = t_1 = 0$*), then* $|\boldsymbol{D}| = p^m$*,* $d_1 = m/p$*,* $d_2 = d_3 = 1/p$ *and* $d_4 = 0$*. If* $q \neq 0$ *and* $p^2 \neq 4q^2$*, let* $r_\pm$ *be the two (distinct) roots of* $r^2 - pr + q^2 = 0$*, so that* $r_+ + r_- = p$ *and* $r_+ r_- = q^2$*, and* $h = r_+ - r_-$ *(a square root of* $p^2 - 4q^2$*; the labeling of the roots and the sign of* h *are fixed in Theorem 2(iii)). Then*

$$|\boldsymbol{D}| = \frac{r_+^{m+1} - r_-^{m+1}}{r_+ - r_-}, \qquad d_3 = \frac{r_+^m - r_-^m}{r_+^{m+1} - r_-^{m+1}}, \qquad d_4 = \frac{(-q)^{m-1}(r_+ - r_-)}{r_+^{m+1} - r_-^{m+1}}, \tag{18}$$

and $d_2 = [1 + q(d_3 + d_4)]/(p + 2q)$*,* $d_1 = [m + 2qd_2]/(p + 2q)$*.*

The closed forms (18) are classical (Meurant, 1992; Hu and O'Connell, 1996; da Fonseca and Petronilho, 2001; Dow, 2003), and the expressions for d_1 and d_2 follow from the row-sum identity $\boldsymbol{D}\mathbf{1}_m = (p + 2q)\mathbf{1}_m - q(\boldsymbol{e}_{1,m} + \boldsymbol{e}_{m,m})$. They break down at and near $p = \pm 2q$, a neighborhood that contains the unit root; Section 4.1 gives the forms used there.

For $\boldsymbol{\Delta} = \boldsymbol{D} + \kappa\mathbf{1}_m\mathbf{1}_m'$, the matrix determinant lemma and the Sherman–Morrison formula give

$$|\boldsymbol{\Delta}| = |\boldsymbol{D}|\,\rho, \qquad \rho = 1 + \kappa d_1, \qquad \boldsymbol{\Delta}^{-1} = \boldsymbol{D}^{-1} - \frac{\kappa}{\rho}\boldsymbol{D}^{-1}\mathbf{1}_m\mathbf{1}_m'\boldsymbol{D}^{-1}, \tag{19}$$

and, with the primitives of Lemma 2, every bilinear form of $\boldsymbol{\Delta}^{-1}$ in the vectors $\mathbf{1}_m$, $\boldsymbol{e}_{1,m}$,

$\boldsymbol{e}_{m,m}$ is an explicit expression in d_1 to d_4:

$$w_1 = \mathbf{1}_m'\boldsymbol{\Delta}^{-1}\mathbf{1}_m = \frac{d_1}{\rho}, \qquad w_2 = \boldsymbol{e}_{1,m}'\boldsymbol{\Delta}^{-1}\mathbf{1}_m = \frac{d_2}{\rho},$$
$$w_3 = \boldsymbol{e}_{1,m}'\boldsymbol{\Delta}^{-1}\boldsymbol{e}_{1,m} = d_3 - \frac{\kappa d_2^2}{\rho}, \qquad w_4 = \boldsymbol{e}_{1,m}'\boldsymbol{\Delta}^{-1}\boldsymbol{e}_{m,m} = d_4 - \frac{\kappa d_2^2}{\rho}, \tag{20}$$

and, by the persymmetry of $\boldsymbol{\Delta}$, $\boldsymbol{e}_{m,m}'\boldsymbol{\Delta}^{-1}\mathbf{1}_m = w_2$ and $\boldsymbol{e}_{m,m}'\boldsymbol{\Delta}^{-1}\boldsymbol{e}_{m,m} = w_3$. Hence

$$\begin{aligned}
\boldsymbol{a}'\boldsymbol{\Delta}^{-1}\boldsymbol{a} &= a_1^2 w_1 + 2a_1a_2w_2 + a_2^2w_3,\\
\boldsymbol{a}'\boldsymbol{\Delta}^{-1}\boldsymbol{b} &= a_1b_1w_1 + (a_1q + a_2b_1)w_2 + a_2qw_4,\\
\boldsymbol{b}'\boldsymbol{\Delta}^{-1}\boldsymbol{b} &= b_1^2w_1 + 2b_1qw_2 + q^2w_3,
\end{aligned}$$

and similarly for g_a, g_b and $\boldsymbol{\beta}_c'\boldsymbol{\Delta}^{-1}\boldsymbol{\beta}_c$ in (16); these are the quantities that Lemma 1 requires. We record the outcome.

Proposition 2 ($O(1)$ evaluation)**.** *For each of the four cases,*

$$\varphi(t_1,t_2) = \varrho\,|\boldsymbol{D}|^{-1/2}\rho^{-1/2}\delta_4^{-1/2}\exp\left(\tfrac{1}{2}\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta} + \eta\right), \tag{21}$$

where $|\boldsymbol{D}|$ and d_1 to d_4 are given by Lemma 2, ρ by (19), δ_4 and $\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta}$ by Lemma 1 through (20) and the coefficients of Table 1. In particular,

$$\mathsf{DK}: \quad |\boldsymbol{K}| = |\boldsymbol{D}|(1-q^2d_3), \qquad \boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta} = q^2z_0^2\Big(d_3 + \frac{q^2d_4^2}{1-q^2d_3}\Big); \tag{22}$$

$$\mathsf{DU}: \quad |\boldsymbol{K}| = |\boldsymbol{D}|\rho\,(1-\boldsymbol{b}'\boldsymbol{\Delta}^{-1}\boldsymbol{b}), \qquad \boldsymbol{b}'\boldsymbol{\Delta}^{-1}\boldsymbol{b} = -\frac{t_1^2}{n^2}w_1 + \frac{2\mathrm{i}t_1q}{n}w_2 + q^2w_3, \tag{23}$$
$$\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta} = \gamma^2\Big[(1-\phi)^2w_1 + \frac{\{(1-\phi)(\mathrm{i}t_1w_1/n + qw_2) - 1\}^2}{1-\boldsymbol{b}'\boldsymbol{\Delta}^{-1}\boldsymbol{b}}\Big];$$

$$\mathsf{SK}: \quad |\boldsymbol{K}| = |\boldsymbol{D}|\big[(p-\phi^2-q^2d_3)(1-q^2d_3) - q^4d_4^2\big]; \tag{24}$$

$$\mathsf{SU}: \quad |\boldsymbol{K}| = |\boldsymbol{D}|\rho\,(\delta_1\delta_3 - \delta_2^2). \tag{25}$$

Every quantity on the right is a scalar obtained from p, q, κ, n and the parameters by a fixed sequence of operations, counting an elementary function or a scalar power at fixed precision as one operation, so $\varphi(t_1,t_2)$ is evaluated with $O(1)$ operations, independent of n.

3.3 Derivatives by Taylor-mode arithmetic

The density (7) requires $\partial\varphi(t_1, t_2)/\partial t_2$ at $t_2 = -rt_1$, and the k-th moment (8) requires $\partial^k M_c(s_1, -s_2)/\partial s_1^k$ at $s_1 = 0$ for fixed $s_2 > 0$. In terms of the characteristic function,

$$M_c(s_1, -s_2) = M(s_1, -s_2 - s_1\tilde{c}) = \varphi\big(-\mathrm{i}s_1,\, \mathrm{i}(s_2 + s_1\tilde{c})\big).$$

Differentiation in s_1 follows this affine path in both transform arguments while s_2 is held fixed. Rather than differentiating the characteristic function by hand — the route taken by Tsui and Ali (1994) and, for the moments, by Sawa (1978) and Vinod and Shenton (1996), and one that leads to long case-specific recursions — we evaluate the assembly of Proposition 2 in *Taylor-mode arithmetic*, a form of automatic differentiation (Griewank and Walther, 2008, Chapter 13). Fix the variable of differentiation and the order K of the highest derivative required: $K = 0$ for the distribution function, which needs only φ (except near $t_1 = 0$, where Section 4.4 needs its derivatives up to order eight in t_1); $K = 1$ in t_2 for the density; and $K = k$ in s_1 for the k-th moment. Every scalar z in the assembly is then carried as the vector $(z^{[0]}, z^{[1]}, \ldots, z^{[K]})$ of its value and its first K derivatives at the point of evaluation, scaled as Taylor coefficients, $z^{[j]} = z^{(j)}/j!$ (a *jet*). Every operation is applied to jets. The product $z = xy$ has $z^{[j]} = \sum_{i=0}^{j} x^{[i]} y^{[j-i]}$, and division, square root, logarithm, exponential and powers have similar recursions of $O(K^2)$ operations (Griewank and Walther, 2008, Table 13.2; Section S3 of the Online Supplement). In exact arithmetic, truncating jets introduces no error in the retained coefficients, because the j-th coefficient of a sum, product or composition depends only on the coefficients of order at most j of its arguments: the output holds the exact derivatives of the analytic formula. In floating point, derivative conditioning, rounding and the truncation of auxiliary Chebyshev expansions affect each derivative separately, and the accuracy of a derivative need not equal that of the function value. The inputs p, q, κ and the border coefficients are affine in the variable of differentiation, so their jets have at most two nonzero entries; the output is the jet of $\log\varphi$, and that of φ follows by one jet exponential. At fixed working precision and fixed auxiliary expansion length, the cost per transform node is independent of n and of order K^3: the elementary jet operations cost $O(K^2)$ each, and the Chebyshev expansions of Section 4.1 are composed with the jet of

their argument by a Horner scheme of K jet multiplications. The number of quadrature nodes in (6), (7) and (8) is a separate matter: it depends on the parameters and the tolerance, and we have no bound on it that holds uniformly in them.

4 Numerical Issues

The closed forms of Lemma 2 are exact in exact arithmetic. Three issues arise in their evaluation, and each produces large errors in a region of interest if left untreated: confluence near the unit root, the branch of the square root, and cancellation in the explosive region. The first and the third are rounding problems; the second arises in exact arithmetic as well. A fourth, more routine one is the scaling of the inversion integrals (Section 4.4).

4.1 Confluence at $\phi = \pm 1$

The expressions (18) are written in the roots $r_\pm$, which coalesce when $p = \pm 2q$. For $q \neq 0$ we define $x = p/(2q)$, so that the roots coalesce when $x = \pm 1$. From (9),

$$p + 2q = (1-\phi)^2 + 2\mathrm{i}[(\phi-1)t_1 - t_2], \qquad p - 2q = (1+\phi)^2 + 2\mathrm{i}[(\phi+1)t_1 - t_2], \quad (26)$$

so $p + 2q$ is small when ϕ is near one and t_1, t_2 are small — the unit-root neighborhood at the origin of the inversion integrals — and vanishes identically along the path $t_2 = 0$ when $\phi = 1$; $p - 2q$ behaves in the same way near $\phi = -1$. As the roots coalesce, $r_+^{m+1} - r_-^{m+1}$ and $r_+ - r_-$ both tend to zero and their ratio loses accuracy; the divisions by $p + 2q$ in d_2 and d_1 compound the loss. Let $\tau = n^2(x+1)$, the natural measure of distance from the confluent point. The computed d_1 is wrong in the ninth digit at $|\tau| \approx 10^{-2}$ and by orders of magnitude at $|\tau| \approx 10^{-6}$ (see Section 7.1). Nothing is singular, however. The leading principal submatrices of $\boldsymbol{D}$ are $\boldsymbol{D}_1, \ldots, \boldsymbol{D}_{m-1}$, and expanding $|\boldsymbol{D}_{k+1}|$ along its last row gives $|\boldsymbol{D}_{k+1}| = p|\boldsymbol{D}_k| - q^2|\boldsymbol{D}_{k-1}|$, $|\boldsymbol{D}_1| = p$, which after the scaling $|\boldsymbol{D}_k|/q^k$ is the three-term recursion of the Chebyshev polynomials of the second kind, $U_k(\cos\theta) = \sin((k+1)\theta)/\sin\theta$, for complex θ, with removable values understood

by continuity. Hence, for $q \neq 0$,

$$|\boldsymbol{D}| = q^m U_m(x), \qquad d_3 = \frac{U_{m-1}(x)}{qU_m(x)}, \qquad d_4 = \frac{(-1)^{m+1}}{qU_m(x)},$$

and the row-sum identities of Lemma 2 with $p + 2q = 2q(x + 1)$ give

$$d_2 = \frac{U_m + U_{m-1} + (-1)^{m+1}}{2q(x+1)\,U_m}, \qquad d_1 = \frac{m(x+1)U_m + U_m + U_{m-1} + (-1)^{m+1}}{2q(x+1)^2\,U_m}, \tag{27}$$

all polynomials evaluated at x. U_m is a polynomial with $U_m(-1) = (-1)^m(m+1) \neq 0$, and the two numerators vanish at $x = -1$, to first and to second order, so the factors $x+1$ cancel. The cure is therefore to evaluate U_m and U_{m-1} near $x = -1$ by their Taylor expansions there. Writing $U_m(x) = \sum_{k=0}^{m} a_k(x+1)^k$, the Chebyshev differential equation gives

$$a_0 = (-1)^m(m+1), \qquad \frac{a_{k+1}}{a_k} = -\frac{(m-k)(m+k+2)}{(2k+3)(k+1)}, \tag{28}$$

with the same recursion at order $m-1$ for U_{m-1}; in these expansions the divisions by $x+1$ and $(x+1)^2$ in (27) are shifts of the coefficients by one and two places, and $U_m(-x) = (-1)^m U_m(x)$ supplies the expansions at $x = +1$. When the expansions are expressed in powers of τ rather than of $x+1$, no coefficient overflows at any m, successive terms fall by a factor of at most $|\tau|/((2k+3)(k+1))$, and twenty terms reach machine precision on $|\tau| < 5/2$ whatever m is. At $x = +1$ the local scaled distance is instead $\tau = n^2(x-1)$; in either neighborhood τ denotes the distance from that neighborhood's confluent point. Our evaluation is accordingly two-regime: the closed forms in the roots when $|\tau| \geq 5/2$ at both confluent points, and the expansions otherwise; on their overlap the two agree to 10^{-12} or better. At large n it also matters that $p \pm 2q$ are formed from the closed forms (26) rather than by subtraction, and $p^2 - 4q^2$ as their product. This is the device of Kan et al. (2026), Section 3.3, which treats only the confluent point $x = -1$; we extend it to $x = +1$, which satisfies $p - 2q = 0$ and arises near $\phi = -1$, and to the unit-root paths on which τ is small over a long stretch or identically zero. The accuracy of the assembled characteristic function is examined in Section 7.1.

4.2 The branch of the complex square root

The inversion formulas require $|\boldsymbol{K}|^{-1/2}$ as a continuous function of t_1 along the path $t_2 = -rt_1$. The argument of $|\boldsymbol{K}|$ grows roughly linearly with n along the path, so if the determinant is assembled first and its principal root taken afterward, the result changes sign at every crossing of the negative real axis, and there are many crossings. We track the modulus and the phase separately: by Proposition 2,

$$|\boldsymbol{K}|^{-1/2} = \exp\Big(-\tfrac{1}{2}\big[\log|\boldsymbol{D}| + \log\rho + \log\delta_4\big]\Big), \tag{29}$$

where the three factors are the determinant of the Toeplitz core $\boldsymbol{D}$, the rank-one factor ρ and the border factor δ_4. Theorem 2 below shows that all three logarithms can be taken continuously. The key is that $\boldsymbol{K}$ and $\boldsymbol{D}$ are accretive (part (i)). For ρ and δ_4 continuity follows directly (part (ii)). For $|\boldsymbol{D}|$, accretivity gives the continuous phase as a sum over the eigenvalues of $\boldsymbol{D}$ (part (i)); parts (iii) and (iv) express the same phase through the roots $r_\pm$ and through the Chebyshev form of Section 4.1, which is how it is computed. Throughout, $\Phi(t_1, t_2)$ denotes the phase of $|\boldsymbol{D}|$: the argument of the determinant of $\boldsymbol{D}$, taken continuously in (t_1, t_2) with $\Phi(0, 0) = 0$, so that $\log|\boldsymbol{D}| = \log|\det \boldsymbol{D}| + \mathrm{i}\Phi$.

Theorem 2 (Continuous branch)**.** *Let $n \geq 3$, with $\phi \in \mathbb{R}$ in cases* SK *and* SU*, and $|\phi| < 1$ in cases* SK *and* SU*.*

(i) For every $(t_1, t_2) \in \mathbb{R}^2$ the matrices $\boldsymbol{K}$ and $\boldsymbol{D}$ of Theorem 1 are accretive, so $|\boldsymbol{K}| \neq 0$, $|\boldsymbol{D}| \neq 0$, and $\Phi(t_1, t_2) = \sum_{j=1}^{m} \operatorname{Arg}\big(p + 2q\cos\frac{j\pi}{n}\big)$, each term lying in $(-\pi/2, \pi/2)$, is the continuous phase of $|\boldsymbol{D}|$, with $\Phi(0, 0) = 0$.

(ii) The rank-one factor $\rho = |\boldsymbol{\Delta}|/|\boldsymbol{D}|$ and the border factor $\delta_4 = |\boldsymbol{K}|/|\boldsymbol{\Delta}|$ never lie on the closed negative real axis, so their principal logarithms are continuous in (t_1, t_2).

(iii) (Roots regime.) Suppose $p + 2q \neq 0$ and $p - 2q \neq 0$, which by (26) excludes only the paths $t_2 = 0$ at $\phi = 1$ and at $\phi = -1$. Let $s_+ = (p + 2q)^{1/2}$ and $s_- = (p - 2q)^{1/2}$ be principal roots, and put

$$r_+ = \frac{(s_+ + s_-)^2}{4}, \qquad r_- = \frac{(s_+ - s_-)^2}{4}, \qquad h = s_+ s_-, \qquad \zeta = \frac{r_-}{r_+}. \tag{30}$$

Then $r_\pm$ are the roots of $r^2 - pr + q^2 = 0$, $r_+ - r_- = h$, $|r_+| > |r_-|$, $\operatorname{Re} r_+ > 0$, $\operatorname{Arg} h \in (-\pi/2, \pi/2)$, so that h is the principal square root of $p^2 - 4q^2$ and $p^2 - 4q^2$ is not on the closed negative real axis, $\operatorname{Re}(1 - \zeta^n) > 0$, *and*

$$\Phi(t_1, t_2) = n \operatorname{Arg}(r_+) + \operatorname{Arg}(1 - \zeta^n) - \operatorname{Arg}(h). \tag{31}$$

(iv) (Confluent regime.) Let $\tau = n^2(x+1)$. On the band $|\tau| < 5/2$ around the confluent point $x = -1$, which arises near the unit root $\phi = 1$, let $\bar{\boldsymbol{D}}$ be $\boldsymbol{D}$ with p replaced by $-2q$, its value at $x = -1$, and $\boldsymbol{\Xi} = \bar{\boldsymbol{D}}^{-1}\boldsymbol{D}$, so that $|\boldsymbol{\Xi}| = U_m(x)/U_m(-1) = (-1)^m U_m(x)/(m+1)$. Then $|\operatorname{Arg}|\boldsymbol{\Xi}|| < 5\pi/12 < \pi/2$, $-q$ is not on the closed negative real axis, and

$$\Phi(t_1, t_2) = m \operatorname{Arg}(-q) + \operatorname{Arg}|\boldsymbol{\Xi}|. \tag{32}$$

The band $n^2|x-1| < 5/2$ around $x = 1$, which arises near $\phi = -1$, is its mirror image under $q \mapsto -q$, which maps x to $-x$: there the same holds with $\bar{\boldsymbol{D}}$ equal to $\boldsymbol{D}$ with p replaced by $2q$, $|\boldsymbol{\Xi}| = U_m(x)/U_m(1) = U_m(x)/(m+1)$, q not on the closed negative real axis, and

$$\Phi(t_1, t_2) = m \operatorname{Arg}(q) + \operatorname{Arg}|\boldsymbol{\Xi}|.$$

Consequently (29), with $\log|\boldsymbol{D}|$ taken as $\log|\det \boldsymbol{D}| + \mathrm{i}\Phi$ and principal logarithms of ρ and δ_4, is the continuous branch of $|\boldsymbol{K}|^{-1/2}$ along every path in the parameter domain of the case; in the fixed-start cases this includes unit roots and explosive coefficients.

The proof is in Appendix A. Parts (i) and (ii) rest on Lemma 3 there. The inverse of an accretive matrix is accretive, so $\operatorname{Re} d_1 > 0$. Since κ is purely imaginary, $\rho = 1 + \kappa d_1$ has a nonzero imaginary part unless $\kappa = 0$, where $\rho = 1$, and so it never lies on the closed negative real axis. The 2×2 Schur complement whose determinant is δ_4 inherits accretivity. In part (iii), $\operatorname{Re}(p + 2q) = (1 - \phi)^2$ vanishes at the unit root and $\operatorname{Re}(p - 2q)$ at $\phi = -1$, so the proof works with the principal roots $s_\pm$ of $p \pm 2q$. The construction (30) also covers $q = 0$, where $x = p/(2q)$ is undefined, and is what the programs compute.

Part (iv) compares the eigenvalues of $\boldsymbol{D}$ with their values at the confluent point: on the band the arguments of the eigenvalues of $\boldsymbol{\Xi}$, the ratios of the two, add up in absolute value to less than $5\pi/12$, so the principal argument of $|\boldsymbol{\Xi}|$ is their sum. The set excluded in (iii) consists of the paths $t_2 = 0$ at $\phi = \pm 1$, on which the confluent regime applies, so the two regimes together cover every real ϕ and every path.

The theorem has two practical consequences. The grid search for discontinuities that Evans and Savin (1984) and Perron (1989) had to perform is unnecessary, the phase (29) being continuous by construction; and the switch between the two regimes of Section 4.1 cannot introduce a jump, because (31) and (32) equal the same function Φ. The same phase serves the density: in the jet arithmetic of Section 3.3 only the constant coefficient of a logarithm carries a branch, the coefficients of order one and above being determined by the derivatives of its argument.

4.3 Cancellation in the explosive region

The third issue is specific to $|\phi| > 1$: nothing degenerates and no branch is in doubt, but the generic determinant assembly of Proposition 2 asks floating-point arithmetic for an $O(1)$ number as the product of a huge number and a tiny one that is itself a difference of $O(1)$ numbers. In case DK at $t_1 = t_2 = 0$, $|\boldsymbol{D}| = (\phi^{2n} - 1)/(\phi^2 - 1)$ is of order ϕ^{2n} while $|\boldsymbol{K}| = 1$, so the border factor δ_4, which in case DK equals $\delta_3 = 1 - q^2 d_3 = (\phi^2 - 1)/(\phi^{2n} - 1)$, is of order ϕ^{-2n}, and the same orders persist along the inversion paths, on which $t_1 = O(|\phi|^{-n})$ and $t_2 = O(\phi^{-2n})$. Computing δ_3 as $1 - q^2 d_3$ incurs a relative error of order $\epsilon_{\mathrm{M}} \phi^{2n}$, where $\epsilon_{\mathrm{M}} = 2^{-52} \approx 2.2 \times 10^{-16}$ is the machine epsilon of double-precision arithmetic. Once ϕ^{2n} reaches 10^{16} this relative error is of order one, so the computed δ_3 has no correct digits; for $\phi = 1.2$ this happens already at $n \approx 100$. The bilinear form $\boldsymbol{\beta}' \boldsymbol{K}^{-1} \boldsymbol{\beta}$ in (22), which divides by $1 - q^2 d_3$, has the same problem, and the obvious alternatives, Gaussian elimination of the bordered matrix in either order and an evaluation through the eigenvalues of the quadratic form whitened by the covariance matrix of the data, do not escape it (Section S6 of the Online Supplement). The remedy is algebraic: the closed forms can be rearranged, once and on paper, so that every small quantity is a sum of small terms.

Proposition 3 (Cancellation-free forms for $|\phi|>1$)**.** *Let $|\phi|>1$, let $r_\pm$, h and $\zeta=r_-/r_+$ be as in Theorem 2(iii), $w=q/r_+$ (so $\zeta=w^2$), and $e_\pm=1-r_\pm$, so that $e_+e_-=1-p+q^2=2\mathrm{i}t_2-t_1^2$. Then*

$$|\boldsymbol{D}|=\frac{r_+^n(1-\zeta^n)}{h},\qquad \delta_{\mathsf{DK}}=1-q^2d_3=\frac{e_--\zeta^ne_+}{1-\zeta^n},\qquad e_-=\frac{2\mathrm{i}t_2-t_1^2}{e_+},\tag{33}$$

and, in case DK*,*

$$\varphi(t_1,t_2)=|\boldsymbol{D}|^{-1/2}\delta_{\mathsf{DK}}^{-1/2}\exp\Big(\frac{z_0^2\,(2\mathrm{i}t_2-t_1^2)}{2\delta_{\mathsf{DK}}}\Big).\tag{34}$$

In case DU*, with $\rho=1+\kappa d_1$ as in (19), let*

$$\begin{aligned}\Theta&=q^2(d_3+d_4)-1=\frac{(-w)^nh-(e_--\zeta^ne_+)}{1-\zeta^n},\\ \Omega&=(1-\phi)qd_2-1=\frac{\mathrm{i}[(1-\phi)t_1+2t_2]+(1-\phi)\Theta}{p+2q}.\end{aligned}\tag{35}$$

Define also $S=1-\boldsymbol{b}'\boldsymbol{\Delta}^{-1}\boldsymbol{b}$, the border factor δ_4 of case DU*, and $\tilde\Omega=(1-\phi)\mathbf{1}_m'\boldsymbol{\Delta}^{-1}\boldsymbol{b}-1$. Then*

$$|\boldsymbol{K}|=|\boldsymbol{D}|\,\rho\,S,\qquad \boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta}=\gamma^2\Big[\frac{(1-\phi)^2d_1}{\rho}+\frac{\tilde\Omega^2}{S}\Big],$$

where

$$\rho S=\rho\,\delta_{\mathsf{DK}}+\frac{2\mathrm{i}}{n}\,qd_2\,(t_1\Omega+t_2qd_2)+\frac{t_1^2}{n^2}\,d_1,\qquad \rho\tilde\Omega=\Omega-\frac{\mathrm{i}}{n}[(1-\phi)t_1+2t_2]\,d_1.\tag{36}$$

Hence, in case DU*,*

$$\varphi(t_1,t_2)=|\boldsymbol{D}|^{-1/2}(\rho S)^{-1/2}\exp\Big\{\frac{\gamma^2}{2}\Big[\frac{(1-\phi)^2d_1}{\rho}+\frac{\tilde\Omega^2}{S}-n\Big]\Big\}.\tag{37}$$

Fix ϕ, z_0 and γ, and let $n\to\infty$. Along the inversion paths, where $t_1=O(|\phi|^{-n})$ and $t_2=O(\phi^{-2n})$, the quantities e_-, ζ^n, δ_{DK}, Θ, Ω, S and $\tilde\Omega$ are of order ϕ^{-2n}, ϕ^{-2n}, ϕ^{-2n}, $|\phi|^{-n}$, $|\phi|^{-n}$, ϕ^{-2n} and $|\phi|^{-n}$, respectively; d_2, ρ, r_+, h and e_+ are $O(1)$, and $d_1=O(n)$. In (33)–(36) no quantity is obtained as the difference of quantities of larger order. The continuous phase of $|\boldsymbol{D}|$ is given by (31), and δ_{DK}, ρ and S never lie on the closed negative real axis, so that in (34)–(37) $\delta_{\mathsf{DK}}^{-1/2}$ and $(\rho S)^{-1/2}=\exp\{-\frac12(\log\rho+\log S)\}$ are evaluated with the principal logarithms of the individual factors. The principal square root of ρS

cannot be used directly, because the product can cross the negative real axis, where that root is discontinuous.

The proof is in Section S6 of the Online Supplement. It rests on the identities, in case DK,

$$|\boldsymbol{K}| = |\boldsymbol{D}_m| - q^2|\boldsymbol{D}_{m-1}| = \frac{r_+^n(e_- - \zeta^n e_+)}{h}, \qquad (-q)^n - |\boldsymbol{K}| = |\boldsymbol{D}|\,\Theta,$$

together with the Schur complement of $\boldsymbol{\Delta}$ in $\boldsymbol{K}$ in case DU. The point of (33) is that e_-, which is $1 - r_-$ with r_- within $O(\phi^{-2n})$ of one, is obtained from the exact product $e_+e_- = 2\mathrm{i}t_2 - t_1^2$ rather than by subtraction. In case DU the bilinear forms $\boldsymbol{\iota}'\boldsymbol{K}^{-1}\boldsymbol{\iota}$ and $\boldsymbol{e}'_{n,n}\boldsymbol{K}^{-1}\boldsymbol{e}_{n,n}$, where $\boldsymbol{\iota} = (\mathbf{1}'_m, 0)'$, are genuinely of order ϕ^{2n}, being variances of explosive partial sums, and (35)–(36) write the small combinations Ω and $\tilde{\Omega}$ in which they enter as sums of terms of the same order. The programs use Proposition 3 whenever $|\phi| > 1$ and $\phi^{2n} > 100$; as $|\phi| \downarrow 1$ the confluent regime takes over on the bands $|\tau| < 5/2$. Against the $O(n)$ backward pivot recursion, which eliminates $\boldsymbol{K}$ one coordinate at a time, carried out in 250-digit arithmetic, on a grid with $\phi \in \{-1.5, 1.1, 1.3, 1.5\}$ and n up to 600 that reaches $\phi^{2n} \approx 2 \times 10^{211}$, the absolute difference in $\log\varphi$ is below 2.5×10^{-13} wherever $|\varphi| > 10^{-17}$. What then limits the explosive region is the range of double precision, but only for $\hat{\phi} - \phi$ itself. Evaluated at t_1 and t_2 as such, (33)–(36) need ϕ^{-2n} to be representable, so the distribution of $\hat{\phi}$ or of $\hat{\phi} - \phi$ can be evaluated as long as $\phi^{2n} \lesssim 10^{300}$, that is, for n up to about 1900 at $\phi = 1.2$. The distribution of the statistic $X_n = |\phi|^n(\hat{\phi} - \phi)/(\phi^2 - 1)$, which is of order one, has no such limit. Its inversion takes place in the scaled arguments $\tau_1 = |\phi|^n t_1$ and $\tau_2 = \phi^{2n} t_2$, and there every small quantity of Proposition 3 is a known power of $|\phi|^{-n}$ times a quantity of order one: e_-, ζ^n, δ_{DK} and S carry the factor ϕ^{-2n}, and Θ, Ω and $\tilde{\Omega}$ the factor $|\phi|^{-n}$. The powers of order n follow from the exact identities

$$\frac{r_+}{\phi^2} = \frac{(1 + \mathrm{i}t_1/\phi)^2}{1 - e_-}, \qquad |\phi|\, w = -\operatorname{sgn}(\phi)\, \frac{1 - e_-}{1 + \mathrm{i}t_1/\phi},$$

through the logarithms of $1+\mathrm{i}t_1/\phi$ and $1-e_-$, which are close to one, so that no number of order one is raised to the power n. Carrying the small quantities divided by their orders, the programs evaluate $\log\varphi(|\phi|^{-n}\tau_1, \phi^{-2n}\tau_2)$ for every n: the factors $|\phi|^{-n}$ that remain multiply terms that are negligible when those factors underflow, and the error does not grow with n. Where both evaluations are possible they agree to about 10^{-11} in $\log\varphi$, and

the scaled one agrees with the 250-digit references to 10^{-13}. Beyond $\phi^{2n} = 10^{300}$, in case DK, where the exact and limiting distributions of X_n differ by $O(|\phi|^{-n})$, the computed distribution function and density of X_n equal the limit (47) of Section 6.2 to 10^{-16} at $n = 5 \times 10^4$ and $n = 5 \times 10^6$ ($\phi = 1.2$, $z_0 = 1$). In case DU, at $\phi = -1.1$ and $\gamma = 0$, n times the difference from the limit (48) at fixed x agrees to four digits for every n from 2000 to 5×10^6, as the $O(n^{-1})$ approach discussed in Section 6.2 requires. The moments of $\hat{\phi} - \phi$ can also be evaluated as long as $\phi^{2n} \lesssim 10^{300}$. They are all of order $\sqrt{n}\,|\phi|^{-n}$ and remain representable, but two precautions are needed. First, at the second scale of the integrand of (39) (Section 5) the tilted spread of $u/\,\mathrm{sd}(u)$ is of order $|\phi|^{-n}$, so the coefficient of order k of the jet in s_1 is of order $|\phi|^{-kn}$, and for $k = 3$ and 4 it underflows once ϕ^{2n} exceeds about 10^{200} and 10^{150}. The programs therefore take the jet in the rescaled variable s_1/ρ, with $\rho = (1 + s_2\mathbb{E}[v])^{1/2}$, and remove the factor ρ^k in logarithms. Second, the moments of the standardized ratio grow like $|\phi|^{(k-1)n}$, so the integral of order k is normalized by the peak of its integrand. Beyond $\phi^{2n} = 10^{300}$ the programs return the moments as NaN, with a warning. The skewness and the excess kurtosis of $\hat{\phi}$ grow like $n^{-1/4}|\phi|^{n/2}$ and $n^{-1/2}|\phi|^n$; in case DU at $\phi = -1.1$ and $\gamma = 0$, as n increases from 1600 to 3600 the ratios of the two to these rates decrease slowly, from 1.408 to 1.403 and from 2.212 to 2.192, and at $\phi^{2n} = 10^{300}$ the skewness and the excess kurtosis are about 10^{74} and 10^{148}. Since $\hat{\phi} - \phi$ is of order $|\phi|^{-n}$, which falls below the spacing of floating-point numbers near ϕ once n exceeds about 200 at $\phi = 1.2$, the programs accept the argument of the distribution function as a value of $\hat{\phi}$, as a value r of $\hat{\phi} - \phi$, as in (6), or, for every n, as a value of X_n.

4.4 The integrand near the origin, its decay, and its scale

The integrand of (6) is a 0/0 form at $t_1 = 0$ with limit $\mathbb{E}[u] - r\mathbb{E}[v]$, and that of (7) tends to $\mathbb{E}[v]$. Closed forms for the limits exist (Evans and Savin, 1984), but a jet is simpler. On an initial interval $[0, t_0]$, the integrand of (6) is replaced by its Taylor polynomial in t_1, whose coefficients are read off a jet of order eight and are exact. The endpoint t_0 is found by halving an initial interval, at most forty times, until the last two retained terms of the polynomial are below one percent of the requested tolerance. For large t_1

the modulus of $\varphi(t_1, -rt_1)$ is $O(t_1^{-\operatorname{rank}(u-rv)/2})$, where the rank is that of the quadratic part of $u - rv$ in nondegenerate Gaussian coordinates. This is an upper bound. Write $u - rv = \boldsymbol{Z}'\boldsymbol{H}\boldsymbol{Z} + \boldsymbol{b}'\boldsymbol{Z} + d$, with $\boldsymbol{Z}$ standard normal and $\boldsymbol{H}$ real symmetric; the component $\boldsymbol{b}_0$ of $\boldsymbol{b}$ in the null space of $\boldsymbol{H}$ contributes the factor $e^{-t_1^2\|\boldsymbol{b}_0\|^2/2}$. The bound is attained when $\boldsymbol{b}_0 = \boldsymbol{0}$, and the decay is Gaussian otherwise. In case DK with $n = 3$, $\phi = 0$ and $z_0 \neq 0$, for instance, $u = z_0e_1 + e_1e_2 + e_2e_3$, and the linear term along the null direction $(1, 0, -1)/\sqrt{2}$ of its quadratic part contributes the factor $e^{-z_0^2t_1^2/4}$ at $r = 0$. The same bound governs the integrand of (7), which is $O(t_1^{-\operatorname{rank}(u-rv)/2})$ as well, so the density is finite when the rank is at least three. The logarithmic assembly (29) never forms the determinant itself, whose modulus can overflow. The natural scale of t_1 is the reciprocal of the standard deviation of $u - rv$, written $1/\operatorname{sd}(u - rv)$; it ranges from $O(n^{-1/2})$ in the stationary case to $O(|\phi|^{-n})$ in the explosive case, where $r = O(|\phi|^{-n})$ at the fixed quantiles of $\hat{\phi} - \phi$. The programs scale t_1 by the proxy $\{\operatorname{Var} u + r^2 \operatorname{Var} v\}^{-1/2}$, which never degenerates, and by the standard deviation of u alone once $\operatorname{Var} v$, of order ϕ^{4n} in the explosive region, overflows. This rescaling helps the adaptive Gauss–Kronrod quadrature (default tolerance 10^{-10}), although it does not bound the total work uniformly across parameters or sample sizes.

Besides the quadrature error, the programs estimate the rounding error of the characteristic function and, for the limits, the truncation error of the asymptotic tail of Section 6.1. They return the total with each value and warn when it exceeds the tolerance; Section S3 of the Online Supplement gives the details. The error is absolute: a quantile r carries the approximate error $|\Delta F|/f(r)$, where ΔF is the error of the distribution function at that quantile. In the tables of Section 7 the error estimates are below 10^{-10}, except at $n \geq 10^6$ (Section 7.1).

5 Exact Moments

With the joint moment generating function available at $O(1)$ cost together with its s_1-derivatives, Sawa's representation (8) becomes a practical formula. Let $L(s_1, s_2) = \log M_c(s_1, -s_2)$, carried as a jet in s_1 at $s_1 = 0$ for fixed $s_2 > 0$; it is computed as the jet

of $\log\varphi$ at the imaginary arguments $(-\mathrm{i}s_1,\, \mathrm{i}(s_2 + s_1\tilde{c}))$. Its coefficients are

$$L_j(s_2) = \frac{1}{j!}\,\frac{\partial^j L(s_1, s_2)}{\partial s_1^j}\bigg|_{s_1=0}, \qquad j = 0, 1, \ldots.$$

They come out of the jet evaluation as numbers, and one jet exponential (Section 3.3) turns the jet of L into that of M_c,

$$\frac{\partial^k M_c(s_1, -s_2)}{\partial s_1^k}\bigg|_{s_1=0} = k!\, e^{L_0} Q_k, \qquad Q_0 = 1, \qquad Q_k = \frac{1}{k}\sum_{j=1}^{k} j\, L_j Q_{k-j}, \quad k \geq 1, \tag{38}$$

where Q_k, the k-th coefficient of the jet of $\exp(L - L_0)$, and $L_0,\ L_1, \ldots$ depend on s_2. Substituting (38) in (8) gives

$$\mathbb{E}[(\hat{\phi} - c)^k] = k \int_0^\infty s_2^{k-1} \exp\{L_0(s_2)\}\, Q_k(s_2)\, \mathrm{d}s_2, \tag{39}$$

a one-dimensional integral of a smooth integrand whose evaluation costs a number of operations per node that is polynomial in k (Section 3.3) and independent of n. At $s_1 = 0$ the matrix $\boldsymbol{K}$ is real and positive definite for every $s_2 \geq 0$, so no branch question arises; the confluence of Section 4.1 does arise at the unit root, where $p + 2q$, evaluated at the arguments $(0, \mathrm{i}s_2)$ of the characteristic function, equals $2s_2 \to 0$, and the Chebyshev regime handles it. When the moment exists, the integrand is $O(s_2^{k-1-(\mathrm{rank}(v)+k)/2})$ for large s_2, except in case DK with $z_0 \neq 0$, where it is exponentially small. The integrand lives on two scales: s_2 of order $1/\mathbb{E}[v]$, set by the largest eigenvalue of the matrix of v, and s_2 of order one over its other eigenvalues; these are a factor n^2 apart at the unit root and ϕ^{2n} apart in the explosive region. We therefore standardize the pair to $(u/\,\mathrm{sd}(u),\, v/\mathbb{E}[v])$, so that the tolerance (10^{-10}) refers to $O(1)$ moments, and integrate adaptively in $\log(s_2\,\mathbb{E}[v])$ over a window that is widened until the omitted tails, estimated from the decay of the integrand at each end, are negligible, so that a distant contribution in the explosive region is not missed (details in the user manual).

Two features deserve comment. First, the centering: at the unit root $\hat{\phi} - 1 = O_p(n^{-1})$, so computing central moments from raw moments is hopeless for large n, whereas (39) with $c = \mathbb{E}[\hat{\phi}]$ (from a first pass with $k = 1$) computes them directly, and the skewness and

excess kurtosis of $n(\hat{\phi}-1)$ at $n=5000$ come with estimated relative errors below 10^{-9}. Second, the scope: earlier exact computations (Sawa, 1978; Tsui and Ali, 1992; Vinod and Shenton, 1996) cover the stationary-start and zero-start cases, while the moments in case DU have been studied by simulation or expansion (Kiviet and Phillips, 2012). Formula (39) covers all four cases. It includes a drift at the unit root, where $\mathbb{E}[v]$ is of order $\gamma^2 n^3$ and the scale $1/\mathbb{E}[v]$ built into the standardization keeps the steep integrand under control. It also includes the explosive region, where the moments of $|\phi|^n(\hat{\phi}-\phi)/(\phi^2-1)$ exist (for every k in case DK with $z_0 \neq 0$, and for k smaller than the rank of v otherwise; Proposition 1) but are dominated by the far tails. At $\phi=1.2$ and $n=100$ its standard deviation is already 1.6×10^4, on its way to a limit whose absolute moments exist only for orders strictly between zero and one.

6 Limiting Distributions

The limiting distribution of $\hat{\phi}$ depends on whether $|\phi|<1$, ϕ is at or near unity, or $|\phi|>1$, and the same machinery evaluates all three. For fixed $|\phi|<1$, $\sqrt{n}(\hat{\phi}-\phi)\Rightarrow\mathcal{N}(0,1-\phi^2)$ in all four cases, by the central limit theorem for martingale differences; the quality of this approximation in finite samples is one of the questions the exact distribution answers. Near the unit root it is poor even in large samples, especially with an intercept: at $\phi=0.99$ and $n=1000$ in case DU with $\gamma=0$, the exact 1%, 50% and 99% quantiles of $\sqrt{n}(\hat{\phi}-\phi)$ are -0.749, -0.109 and 0.168, against -0.328, 0 and 0.328 for the limit $\mathcal{N}(0,1-\phi^2)$. In the other two regions, ϕ at or near unity and $|\phi|>1$, the limits are nonstandard; the next two subsections present them and show how they are evaluated.

6.1 The unit root and its neighborhood

In this subsection $c\in\mathbb{R}$ is the fixed local-to-unity parameter, rather than the moment-centering constant of Section 5. Let $\phi=1+c/n$, W be a standard Brownian motion, and $J_c(r)=\int_0^r e^{(r-s)c}\mathrm{d}W(s)$ be the Ornstein–Uhlenbeck process it generates. The initial value enters the limits through the scaled start $\gamma_0=y_0/(\sqrt{n}\,\sigma)$, which we keep distinct from the finite-sample parameter γ of (5). In case DK, where $z_0=\sqrt{n}\,\gamma_0$, Phillips (1987)

and Perron (1991a) show that $n(\hat{\phi}-\phi) \Rightarrow A/B$ with

$$A = \gamma_0 \int_0^1 e^{cr} \mathrm{d}W(r) + \int_0^1 J_c(r)\, \mathrm{d}W(r), \tag{40}$$

$$B = \gamma_0^2 \frac{e^{2c}-1}{2c} + 2\gamma_0 \int_0^1 e^{cr} J_c(r) \mathrm{d}r + \int_0^1 J_c^2(r) \mathrm{d}r, \tag{41}$$

and in case SU with $\alpha = b\sigma/\sqrt{n}$ for a fixed real drift parameter b, $n(\hat{\phi}-\phi) \Rightarrow \hat{A}/\hat{B}$, where

$$\hat{A} = \int_0^1 X(r)\, \mathrm{d}W(r) - W(1) \int_0^1 X(r)\, \mathrm{d}r, \qquad \hat{B} = \int_0^1 X^2(r)\, \mathrm{d}r - \left(\int_0^1 X(r)\, \mathrm{d}r \right)^2, \tag{42}$$

with $X(r) = J_c(r) + \varkappa H_c(r)$, $H_c(r) = (e^{cr}-1)/c$ ($= r$ at $c = 0$) and $\varkappa = b + c\gamma_0$. The limiting process of the standardized data is $\gamma_0 + \varkappa H_c(r) + J_c(r)$, and demeaning removes the constant γ_0, so the limit depends on (b, γ_0) only through $\varkappa$ (Zhou and Yu, 2015, Theorem 2.1, where it is written in (b, γ_0); Section S1 of the Online Supplement gives the reduction). The representation (42) is continuous in c: at $c = 0$, $H_0(r) = r$ and $X(r) = W(r) + br$, the random walk with drift of Haldrup and Hylleberg (1995), and with $\varkappa = 0$, $c = 0$ it gives the Dickey–Fuller distributions. The stationary cases require $c < 0$, and in case SU we may take $\alpha = 0$, so that $b = 0$, because $\hat{\phi}$ is invariant to the mean. The initial observation is then $y_0 \sim \mathcal{N}(0, \sigma^2/(1-\phi^2))$ with $1-\phi^2 \approx -2c/n$, so $y_0/(\sqrt{n}\sigma) \Rightarrow \gamma_0 \sim \mathcal{N}(0, -1/(2c))$, independent of W; this is the stationary initial condition of Elliott (1999) and Müller and Elliott (2003). The limits of cases SK and SU are (40)–(41) and (42) with this random γ_0, and with $\varkappa = c\gamma_0$ in case SU.

The joint characteristic functions are known in closed form in all four cases.

Proposition 4 (Limiting characteristic functions)**.** *For* $(t_1, t_2) \in \mathbb{R}^2$, *let* $\omega = c + \mathrm{i}t_1$, $\theta^2 = 2\mathrm{i}t_2 - 2\mathrm{i}ct_1 - c^2$, $\tau_0 = t_1^2 - 2\mathrm{i}t_2$, $\mathcal{C} = \cos\theta$, $\mathcal{S} = \sin\theta/\theta$,

$$\bar{d}_2 = \frac{\tan(\theta/2)}{\theta}, \qquad \bar{d}_1 = \frac{2\bar{d}_2 - 1}{\theta^2}, \qquad \rho_\infty = 1 + (\theta^2 + c^2)\bar{d}_1,$$

and

$$\mathcal{S}_\infty = \theta \cot\theta - \omega + \frac{(\theta^2 + c^2)\bar{d}_2^2 + 2\mathrm{i}t_1\bar{d}_2 + t_1^2\bar{d}_1}{\rho_\infty},$$

all even functions of θ and hence functions of θ^2, analytic on the parameter domain (removable singularities being filled in by continuity). Then

$$\mathsf{DK}: \ \mathbb{E}[e^{\mathrm{i}t_1 A+\mathrm{i}t_2 B}] = e^{-\omega/2}\left(\mathcal{C}-\omega\mathcal{S}\right)^{-1/2}\exp\Big(-\frac{\gamma_0^2\tau_0\mathcal{S}}{2(\mathcal{C}-\omega\mathcal{S})}\Big), \tag{43}$$

$$\mathsf{DU}: \ \mathbb{E}[e^{\mathrm{i}t_1 \hat{A}+\mathrm{i}t_2 \hat{B}}] = e^{-\omega/2}\left(\mathcal{S}\rho_\infty\mathcal{S}_\infty\right)^{-1/2}\exp\Big(-\frac{\varkappa^2\tau_0\bar{d}_1}{2\rho_\infty\mathcal{S}_\infty}\Big), \tag{44}$$

$$\mathsf{SK}: \ \mathbb{E}[e^{\mathrm{i}t_1 A+\mathrm{i}t_2 B}] = e^{-\omega/2}\left(\mathcal{C}-\omega\mathcal{S}\right)^{-1/2}\Big(1-\frac{\tau_0\mathcal{S}}{2c(\mathcal{C}-\omega\mathcal{S})}\Big)^{-1/2}, \quad c<0, \tag{45}$$

$$\mathsf{SU}: \ \mathbb{E}[e^{\mathrm{i}t_1 \hat{A}+\mathrm{i}t_2 \hat{B}}] = e^{-\omega/2}\left(\mathcal{S}\rho_\infty\mathcal{S}_\infty\right)^{-1/2}\Big(1-\frac{c\,\tau_0\bar{d}_1}{2\rho_\infty\mathcal{S}_\infty}\Big)^{-1/2}, \quad c<0, \tag{46}$$

where the square roots are the branches continuous in (t_1, t_2) for which the right-hand sides equal one at the origin (for instance, $\mathcal{C}-\omega\mathcal{S} = e^{-c}$ at $t_1 = t_2 = 0$ and its root is $e^{-c/2}$, which the factor $e^{-\omega/2}$ cancels).

Formula (43) is Perron (1991a), Theorem 1, and (44) is a rearrangement of the characteristic function implied by Nabeya and Tanaka (1990), Nabeya and Sørensen (1994), Theorem 1, and Zhou and Yu (2015); (45) and (46) are the stationary-start characteristic functions of Tanaka (1990), Section 4.2, and Perron (1991b) (case SK) and of Nabeya and Sørensen (1994), Corollary 1 (case SU), written in the same form. The factors of (44) and (46) are those of the finite-sample assembly (21) in the limit. Under $(t_1, t_2) \mapsto (t_1/n, t_2/n^2)$ and $\phi = 1 + c/n$, $|\boldsymbol{D}|/n \to e^{\omega}\mathcal{S}$, the rank-one factor $\rho \to \rho_\infty$ and, in case DU, $n\delta_3 \to \mathcal{S}_\infty$, with $\bar{d}_1$ and $\bar{d}_2$ the limits of d_1/n^3 and d_2/n. In case DK likewise $n\delta_3 \to \theta\cot\theta - \omega$, and

$$e^{\omega}\mathcal{S}(\theta\cot\theta-\omega) = e^{\omega}(\mathcal{C}-\omega\mathcal{S})$$

(Section S1.4 of the Online Supplement). The stationary forms follow by integrating out γ_0 analytically, as in Nabeya and Sørensen (1994): the exponent in (43) is $\gamma_0^2\Lambda$, with $\Lambda = -\tau_0\mathcal{S}/(2(\mathcal{C}-\omega\mathcal{S}))$, and

$$\mathbb{E}[e^{\gamma_0^2\Lambda}] = (1-2\Lambda\operatorname{Var}\gamma_0)^{-1/2}, \qquad \operatorname{Var}\gamma_0 = -\frac{1}{2c}$$

(the derivations are in the proof of Lemma S1.2 of the Online Supplement). All the functions of θ in (43)–(46) are even, so the choice of the root θ of θ^2 is immaterial: they

are evaluated by power series in θ^2 when $|\theta^2| < 4$ and otherwise from closed forms in the principal root $\vartheta = (-\theta^2)^{1/2}$ and $e^{-\vartheta}$, arranged so that no $O(1)$ quantity is a difference of quantities of order e^{ϑ}.

Each square root is evaluated on its continuous branch by writing its argument as a product of factors whose arguments are known (Lemmas S1.1–S1.4 of the Online Supplement). The last factors of (45) and (46), which come from integrating out γ_0, have real part at least one, so they are not on the closed negative real axis. The factor $\mathcal{C} - \omega\mathcal{S}$ is $\cos\theta$ times a factor that is not on the closed negative real axis either, and $\cos\theta$ circles the origin many times along an inversion path. With ϑ as above, however, $\cos\theta = \frac{1}{2}e^{\vartheta}(1 + e^{-2\vartheta})$, where e^{ϑ} has the continuous argument $\operatorname{Im}\vartheta$ and $1 + e^{-2\vartheta}$ lies in the closed right half-plane. The continuous argument of $\cos\theta$ is therefore $\operatorname{Im}\vartheta + \operatorname{Arg}(1 + e^{-2\vartheta})$. This replaces the sign-switching rule of Evans and Savin (1981b) and Perron (1989). In cases DU and SU, $\mathcal{S} = \mathcal{W}\mathcal{S}_h^2$ with $\mathcal{W} = (\theta/2)\cot(\theta/2)$ and $\mathcal{S}_h = \sin(\theta/2)/(\theta/2)$, so

$$(\mathcal{S}\rho_\infty\mathcal{S}_\infty)^{-1/2} = \mathcal{S}_h^{-1}(\mathcal{W}\rho_\infty\mathcal{S}_\infty)^{-1/2}.$$

Here $\mathcal{S}_h$ is single-valued and positive at the origin, and $\mathcal{W}$, ρ_∞ and $\mathcal{S}_\infty$ each stay off the closed negative real axis, so the sum of their principal arguments is the continuous phase of $\mathcal{W}\rho_\infty\mathcal{S}_\infty$.

With the characteristic functions in hand, the distribution functions, densities, quantiles and moments of the four limits follow from (6), (7) and (8) exactly as in the finite-sample case, with θ^2, ω and τ_0 as jets. These limits have moments of all orders (Online Supplement, Section S1.5), in contrast with the limit under a fixed $|\phi| > 1$ in Section 6.2, whose absolute moments exist only for orders strictly between zero and one. For the limits, the inversion integrals need one extra device. In finite samples the integrands of (6) and (7) decay at least like $t_1^{-\operatorname{rank}(u-rv)/2}$ (Section 4.4), fast enough for adaptive quadrature. In the limit, at $r = -c$ the variable $A + cB$ is of χ^2 type (Section S2 of the Online Supplement), so its characteristic function decays only like $t_1^{-1/2}$ in case DK and t_1^{-1} in the other cases. For r near $-c$ the integrands inherit this slow decay up to t_1 of order $1/|r+c|$, and adaptive quadrature on $[0, \infty)$ then stops with an error of order 10^{-5}, however small the tolerance. We therefore integrate over $[0, \bar{t}]$, with $\bar{t} > 0$ the integration cutoff, and

complete the integral by repeated integration by parts: for every integer $M \geq 1$,

$$\int_{\bar{t}}^{\infty} \psi(t_1) e^{-\mathrm{i}w_0 t_1} \mathrm{d}t_1 = e^{-\mathrm{i}w_0 \bar{t}} \sum_{k=0}^{M-1} \frac{\psi^{(k)}(\bar{t})}{(\mathrm{i}w_0)^{k+1}} + \frac{1}{(\mathrm{i}w_0)^M} \int_{\bar{t}}^{\infty} \psi^{(M)}(t_1) e^{-\mathrm{i}w_0 t_1} \mathrm{d}t_1,$$

where w_0 is the slope of the dominant linear phase of the characteristic function (1/2, plus $\gamma_0^2/2$ in case DK) and ψ is the slowly varying remainder of the integrand. The identity holds whenever $\psi^{(k)}(t_1) \to 0$ as $t_1 \to \infty$ for $k < M$ and $\psi^{(M)}$ is integrable on $[\bar{t}, \infty)$, which is the case here because ψ and its derivatives decay like powers of t_1; it is exact for every M, while the sum alone is an asymptotic, not a convergent, expansion. The derivatives $\psi^{(k)}(\bar{t})$ are read off a jet of order eight at $\bar{t}$; we use $M = 8$ for the distribution function and $M = 7$ for the density, whose integrand consumes one order of the jet, and drop the remainder. We truncate at $\bar{t} = 400/w_0$ ($\bar{t} = 800$ when $w_0 = 1/2$), and double $\bar{t}$, at most five times, while the first omitted term exceeds one tenth of the tolerance. Since ψ varies on the scale of t_1 itself, $\psi^{(k)}(\bar{t})$ is of order $k!\,\psi(\bar{t})/\bar{t}^k$, so the k-th term of the sum is smaller than the first by a factor of order $k!/(w_0\bar{t})^k = k!/400^k$. We add the first omitted term ($k = M$) to the error estimate, as a diagnostic of the size of the remainder rather than a bound on it; it is below 10^{-11} in every case we have examined. At $r = -c$ itself the distribution function is evaluated from the closed forms of Section S2 of the Online Supplement. At $c = 0$, with $\gamma_0 = 0$ in case DK and $\varkappa = 0$ in case DU, the limits are the Dickey–Fuller distributions, and we recover $\mathbb{P}[A/B < 0] = 2\Phi_{\mathcal{N}}(1) - 1 = 0.68268949$ and $\mathbb{P}[\hat{A}/\hat{B} < 0] = 0.95756559$ (Evans and Savin, 1981b), $\mathbb{E}[A/B] = -1.78143$ and $\mathbb{E}[\hat{A}/\hat{B}] = -5.37913$ (Gonzalo and Pitarakis, 1998; Nabeya, 1999), to all digits shown; the inversion integral with the asymptotic tail reproduces the closed forms at $r = -c$ to 10^{-10} in all four cases.

6.2 The explosive region

When $|\phi| > 1$ and $y_0 = z_0\sigma$ is fixed (case DK), White (1958) and Anderson (1959) show that

$$\frac{|\phi|^n}{\phi^2 - 1}(\hat{\phi} - \phi) \Rightarrow \frac{Z_1}{Z_2}, \qquad Z_1 \sim \mathcal{N}(0,1), \quad Z_2 \sim \mathcal{N}\big(z_0\sqrt{\phi^2 - 1}, 1\big), \quad Z_1 \perp Z_2, \tag{47}$$

a standard Cauchy variable when $z_0 = 0$. The limit is the same with the signed power ϕ^n in place of $|\phi|^n$, because the two normalizations differ by the sign $(-1)^n$ when $\phi < -1$, and Z_1 is symmetric and independent of Z_2. For any z_0, put $\lambda = z_0^2(\phi^2 - 1)/2$ and define the error function $\operatorname{erf}(x) = 2\pi^{-1/2} \int_0^x e^{-t^2}\, \mathrm{d}t$. The limiting density of Z_1/Z_2 at $x \in \mathbb{R}$ is

$$f_\infty(x) = \frac{e^{-\lambda} + \sqrt{\pi\lambda/(1+x^2)}\, e^{-\lambda x^2/(1+x^2)} \operatorname{erf}\bigl(\sqrt{\lambda/(1+x^2)}\bigr)}{\pi(1+x^2)}.$$

The limit has no absolute moments of order $k \geq 1$, while the finite-sample distribution has all of them when $z_0 \neq 0$. So the moments of $\hat{\phi}$ are finite at every n, but the absolute moments of order $k \geq 1$ of the normalized statistic $|\phi|^n(\hat{\phi} - \phi)/(\phi^2 - 1)$ diverge as n grows. This follows from weak convergence to the limit, applied to bounded truncations of those moments. Signed odd moments need not diverge, because positive and negative contributions can cancel. The exact distribution shows how the heavy tails form (Section 7.4).

The exact finite-sample analysis requires nothing new analytically, and numerically it requires two things. First, the integration variable is rescaled by $\operatorname{sd}(u - rv)$, which is of order $|\phi|^n$ (Section 4.4). Second, it requires the cancellation-free forms of Proposition 3. With these the programs reach $\phi^{2n} \approx 10^{300}$ for $\hat{\phi} - \phi$ itself, and, in scaled arguments, every n for the normalized statistic, at the same $O(1)$ cost (Section 4.3).

With an intercept (case DU) the limit is again a ratio of independent normals: Wang and Yu (2015, Theorem 2.3(b)) show that

$$\frac{|\phi|^n}{\phi^2 - 1}(\hat{\phi} - \phi) \Rightarrow \frac{Z_1}{Z_2}, \qquad Z_1 \sim \mathcal{N}(0,1), \quad Z_2 \sim \mathcal{N}\Bigl(\frac{\gamma(\phi+1)}{\sqrt{\phi^2-1}}, 1\Bigr), \quad Z_1 \perp Z_2, \tag{48}$$

which is (47) with z_0 replaced by $\gamma/(\phi - 1)$, and a standard Cauchy variable when $\gamma = 0$.

In finite samples, however, the limit (48) is a poor approximation in case DU, unlike (47) in case DK (Figure 2). The demeaning enters at order n^{-1}, so the error of the limit is of order n^{-1} in case DU, against $|\phi|^{-n}$ in case DK: at $\phi = 1.175$ with $\gamma = 0$ the maximal error in the distribution function of the normalized statistic is 0.11, 0.022 and 0.0053 at $n = 50$, 200 and 800 (Table S10 of the Online Supplement).

A far better approximation is available in closed form (Proposition S5.1 of the Online

Supplement): a modification of the ratio (48) whose error is $O(n|\phi|^{-n/5})$ uniformly in x, smaller than any power of n^{-1}. At $\phi = 1.175$, $\gamma = 0$, and $n = 200$, our proposed approximate distribution function agrees with the exact one to 10^{-11}.

7 Numerical Results

This section documents the accuracy and speed of the algorithms and reports exact results that previously required simulation. All computations were done on a laptop with an Intel Core Ultra 9 185H processor, Windows 11, MATLAB R2025b with six computational threads, with the programs of the distributed version of November 2026, except the high-precision references, which use the Python library `mpmath`.

7.1 Accuracy and speed

Table 2 compares the $O(1)$ evaluation of the characteristic function with a dense Gaussian quadratic-form evaluation. Let $\bar{n} = n$ for fixed starts and $\bar{n} = n + 1$ for stationary starts, and let $\lambda_j(r)$, $j = 1, \ldots, \bar{n}$, be the eigenvalues of the symmetric matrix representing the quadratic part of $u - rv$ in independent standard normal coordinates. The determinant factor of $\varphi(t_1, -rt_1)$ is $\prod_{j=1}^{\bar{n}}(1 - 2it_1\lambda_j(r))^{-1/2}$; the Gaussian linear and constant terms supply the noncentral exponential when the start or drift is nonzero. This eigenvalue construction gives the continuous branch along the inversion paths $t_2 = -rt_1$, at $n = 60$ and 200, for four values of the threshold r of (6), the mean of $\hat{\phi} - \phi$ minus three and one and plus one half and two of its standard deviations (both approximated by the ratios of the moments of u and v), so that the paths cover the body and both tails of the distribution. For $\phi = 1.1$ the grid and the reference are modified as described in the notes to the table: in double precision both the eigenvalue evaluation and the generic determinant assembly of Proposition 2 lose accuracy at $n = 200$, where $\phi^{2n} \approx 3.6 \times 10^{16}$. The table shows the cancellation-free forms of Proposition 3 to be as accurate as the rest. The maximum absolute differences are generally of order 10^{-14} to 10^{-13}, with a largest recorded difference of 9.4×10^{-12} (case DU, $\gamma = 1$, $n = 200$); these are also the relative accuracies where the function is not small. Near the unit root the accuracy degrades slowly with n, roughly like $n\epsilon_{\mathrm{M}}$ at large n. The error estimates returned with $\mathbb{P}[n(\hat{\phi} - 1) < -8]$

Table 2: Accuracy of the $O(1)$ evaluation of the characteristic function

Case	ϕ	parameter	$n = 60$	$n = 200$
DK	0.5	$z_0 = 0$	1.1×10^{-14}	3.2×10^{-14}
DK	1	$z_0 = 0$	1.6×10^{-14}	5.9×10^{-14}
DK	1	$z_0 = 1$	1.8×10^{-14}	6.1×10^{-14}
DK	$1.1^{\dagger}$	$z_0 = 0.5$	2.0×10^{-14}	2.2×10^{-14}
DK	-1	$z_0 = 0$	1.8×10^{-14}	5.6×10^{-14}
DU	0.9	$\gamma = 0$	1.2×10^{-14}	3.5×10^{-14}
DU	1	$\gamma = 0$	2.0×10^{-14}	7.6×10^{-14}
DU	1	$\gamma = 1$	9.1×10^{-13}	9.4×10^{-12}
DU	$1.1^{\dagger}$	$\gamma = 0.5$	3.0×10^{-14}	6.8×10^{-14}
DU	-1	$\gamma = 0$	1.6×10^{-14}	6.6×10^{-14}
SK	0.95	–	1.3×10^{-14}	3.7×10^{-14}
SK	0.999	–	2.7×10^{-13}	2.8×10^{-13}
SU	0.95	–	1.2×10^{-14}	3.1×10^{-14}
SU	0.999	–	4.2×10^{-13}	3.7×10^{-13}

Notes: Maximum absolute difference between the closed-form evaluation of $\varphi(t_1, -rt_1)$ (Proposition 2, and Proposition 3 in the rows $\phi = 1.1$, with the phase of Theorem 2) and a dense eigenvalue-based evaluation, over $t_1 \in [10^{-5}, 10^3]$ (2000 logarithmically spaced points) and four thresholds r, the approximate mean of $\hat{\phi}-\phi$ plus -3, -1, 0.5 and 2 approximate standard deviations (for $\phi = 1.1$, $r = \pm 0.1$ and ± 2 times $\phi^{-n}(\phi^2-1)$, the scale of $\hat{\phi} - \phi$ there). †For $\phi = 1.1$ the points are $t_1 = s/\operatorname{sd}(u - rv)$, $s \in [10^{-5}, 10^3]$, since $\operatorname{sd}(u - rv)$ is of order $|\phi|^n$; the reference is the eigenvalue evaluation carried out in 40-digit arithmetic (`mpmath`), since in double precision it is itself inaccurate at $n = 200$ (its covariance matrix has entries of order ϕ^{2n}).

in case DU with $\phi = 1$ and $\gamma = 0$ are 2×10^{-12} at $n = 10^2$, 1.5×10^{-10} at $n = 10^6$ and 5×10^{-8} at $n = 10^8$, and the values approach the Dickey–Fuller limit 0.21811104 at the rate $1.06/n$, as they should. Table S4 in the Online Supplement isolates the confluence problem. At $t_1 = t_2 = 0$ and $n = 120$, the closed form in the roots loses the ninth digit of d_1 at $\phi = 0.999$ and has relative error 7.6×10^5 at $\phi = 0.999999$, whereas the Chebyshev expansion is at machine precision throughout. Figure S1 there shows the same along a path at $n = 600$ and $\phi = 1 - 10^{-7}$.

In the benchmark of Table 3, a value of the distribution function costs about five milliseconds (three to four milliseconds in case SU), a density value about a hundredth of

Table 3: Cost against the sample size (seconds)

	$O(1)$ algorithm					alternatives	
n	cdf	pdf	quantile	four moments	cdf, SU	$O(n)$ cdf	eigenvalues
10^2	0.0034	0.0056	0.049	0.025	0.0033	0.08	0.02
10^3	0.0041	0.0084	0.058	0.024	0.0040	0.41	0.25
10^4	0.0049	0.0107	0.061	0.028	0.0029	3.34	158.15
10^5	0.0041	0.0089	0.053	0.029	0.0040	25.45	–
10^6	0.0053	0.0100	0.088	0.028	0.0028	169.54	–

Notes: Case DU, $\phi = 1$, $\gamma = 0$: one value of $\mathbb{P}[n(\hat{\phi} - 1) < -8]$, one value of the density at the same point, the 5% quantile, and $\mathbb{E}[(\hat{\phi} - 1)^k]$, $k = 1, \ldots, 4$ (four one-dimensional quadratures). Column "cdf, SU": $\mathbb{P}[\hat{\phi} < 0.95]$ in case SU with $\phi = 0.95$. Each of these entries is the average of five runs after a warm-up call, with the default tolerances. "$O(n)$ cdf": the same value of $\mathbb{P}[n(\hat{\phi} - 1) < -8]$ with the characteristic function evaluated by the $O(n)$ backward pivot recursion instead of the $O(1)$ assembly (one run). "Eigenvalues": one eigendecomposition of the matrix of $u - rv$ in independent standard normal coordinates, which the eigenvalue approach needs for each value of r (one run; not attempted beyond $n = 10^4$).

a second, a quantile under a tenth of a second and four moments about three hundredths, with no trend in n between 10^2 and 10^6. These are timings of one configuration on one machine: the number of quadrature nodes, and with it the time, depends on the case and the parameters, and the number of operations per transform evaluation, counting an elementary function or a scalar power at fixed precision as one operation, is what does not grow with n. The table also gives two alternatives. The first computes the same value by an $O(n)$ pivot recursion of the kind used by Evans and Savin (1984) and Tsui and Ali (1994): Gaussian elimination of the matrix $\boldsymbol{K}$ of Theorem 1 from its last coordinate backwards, each pivot being the Schur complement of an accretive matrix (Lemma 3), so that the sum of the principal arguments of the pivots is the continuous phase of $|\boldsymbol{K}|$ with no branch condition; it uses none of the closed forms of Section 3.2. The recursion costs 0.08 seconds at $n = 10^2$ and grows roughly linearly, to three minutes at $n = 10^6$. It agrees with the $O(1)$ value to ten digits up to $n = 10^4$, but only to 3×10^{-8} at $n = 10^6$, where its rounding errors accumulate over the n steps. The eigenvalue approach costs $O(n^3)$ per value of r: one eigendecomposition takes a quarter of a second at $n = 10^3$ and about 2.6 minutes at $n = 10^4$.

Table 4: Exact quantiles of $n(\hat{\phi}-1)$ at $\phi=1$, fixed start ($z_0=0$ in DK, $\gamma=0$ in DU)

n	0.01	0.025	0.05	0.10	0.50	0.90	0.95	0.975	0.99
Case DK*: no intercept*									
25	−11.920	−9.363	−7.371	−5.350	−0.814	1.012	1.411	1.789	2.294
50	−12.756	−9.876	−7.692	−5.526	−0.833	0.968	1.345	1.696	2.156
100	−13.212	−10.151	−7.862	−5.618	−0.843	0.948	1.315	1.653	2.093
250	−13.499	−10.323	−7.967	−5.675	−0.849	0.936	1.297	1.628	2.056
500	−13.596	−10.381	−8.003	−5.694	−0.851	0.932	1.291	1.620	2.044
1000	−13.646	−10.411	−8.021	−5.704	−0.852	0.930	1.288	1.616	2.038
∞	−13.695	−10.440	−8.039	−5.714	−0.853	0.928	1.285	1.612	2.033
Case DU*: intercept estimated*									
25	−16.631	−14.105	−12.060	−9.886	−4.084	−0.727	−0.0005	0.606	1.323
50	−18.450	−15.401	−13.008	−10.528	−4.216	−0.787	−0.074	0.509	1.181
100	−19.488	−16.128	−13.532	−10.879	−4.285	−0.817	−0.109	0.464	1.116
250	−20.158	−16.593	−13.864	−11.099	−4.328	−0.834	−0.129	0.438	1.079
500	−20.390	−16.753	−13.978	−11.174	−4.342	−0.840	−0.136	0.429	1.066
1000	−20.507	−16.834	−14.036	−11.212	−4.350	−0.843	−0.139	0.425	1.060
∞	−20.626	−16.915	−14.094	−11.251	−4.357	−0.845	−0.143	0.421	1.054

Notes: Exact quantiles x_p with $\mathbb{P}[n(\hat{\phi}-1)<x_p]=p$, for the regression of y_t on y_{t-1} (case DK, $z_0=0$) or on a constant and y_{t-1} (case DU, $\gamma=0$) over $t=1,\ldots,n$; the last row of each panel is the Dickey–Fuller limit computed from Proposition 4.

7.2 Exact critical values at the unit root

Table 4 gives the exact quantiles of $n(\hat{\phi}-1)$ when $\phi=1$, with $z_0=0$ in case DK (no intercept) and $\gamma=0$ in case DU (intercept estimated), at $n=25$, 50, 100, 250, 500 and 1000, a set that includes the sample sizes of Table 10.A.1 of Fuller (1996), together with the limiting quantiles from Section 6.1. Fuller's table, constructed by simulation (50,000 replications at each n, smoothed), carries errors of about ±0.1 in the tails, and the exact values confirm it to that precision: at $n=25$ the exact quantiles without an intercept agree with Fuller's to within 0.08. With an intercept, Fuller's entry at sample size n comes from $n-1$ regression observations scaled by n, so it is not directly comparable with our quantile for n observations; after adjustment for the observation count and the scaling, it agrees with the exact distribution to his stated precision (Online Supplement, Section S4). The limiting 5% points, −8.039 and −14.094, agree with the response surfaces of MacKinnon (1996) to the three digits he reports, and the probabilities and percentage points of Evans and Savin (1984) for case DU are reproduced; Section S4 of the Online

Supplement gives the details of these comparisons. Since each entry costs a fraction of a second, there is no longer any reason to interpolate in n; Tables S2 and S3 of the Online Supplement give further sample sizes and probabilities.

7.3 Comparison with existing tables and software

Table 5: Simulated tables versus the exact distribution: critical values of the Dickey–Fuller coefficient statistic $T_e(\hat{\phi}-1)$, $T_e = T-1$

	no intercept (`AR`; exact: DK, $n=T$)						intercept (`ARD`; exact: DU, $n=T-1$)					
	1%			5%			1%			5%		
T	table	exact	size	table	exact	size	table	exact	size	table	exact	size
25	−11.841	−11.443	0.86	−7.332	−7.076	4.56	−16.469	−16.494	1.01	−11.969	−11.987	5.03
50	−12.737	−12.501	0.93	−7.685	−7.538	4.77	−18.413	−18.409	1.00	−12.981	−12.987	5.01
100	−13.217	−13.080	0.96	−7.862	−7.783	4.88	−19.480	−19.477	1.00	−13.513	−13.527	5.02
250	−13.492	−13.445	0.99	−7.962	−7.936	4.96	−20.138	−20.157	1.00	−13.850	−13.863	5.02
500	−13.597	−13.569	0.99	−8.001	−7.987	4.98	−20.401	−20.390	1.00	−13.992	−13.978	4.98
1000	−13.664	−13.632	0.99	−8.024	−8.013	4.98	−20.507	−20.507	1.00	−14.038	−14.036	5.00
5000	−13.678	−13.683	1.00	−8.033	−8.034	5.00	−20.556	−20.602	1.01	−14.061	−14.082	5.03

Notes: "table": the critical value returned by `adftest` (MATLAB R2025b, `'Test','t2','Lags',0`), interpolated from tables simulated with 5×10^6 replications; "exact": the quantile from (6); "size": the exact rejection probability, in percent, of the test that uses the tabulated critical value at the stated nominal level.

Standard software obtains Dickey–Fuller critical values and p-values from simulation. According to its documentation, `adftest` in the MATLAB Econometrics Toolbox (R2025b) tabulates the quantiles of each statistic from five million Gaussian replications for a grid of sample sizes and interpolates. In R, `tseries::adf.test`, `urca::ur.df` and `fUnitRoots` use the tables of Banerjee et al. (1993), of Fuller (1996) and Hamilton (1994), or the response surfaces of MacKinnon (1996). Table 5 compares `adftest` with the exact distribution for the Dickey–Fuller coefficient statistic, which `adftest` calls `t2`, with no augmentation lags and for the models without (`AR`) and with (`ARD`) an intercept. The statistic is $T_e(\hat{\phi}-1)$, where T is the observed-series length and $T_e = T-1$ is the number of observations in the regression of y_t on y_{t-1} over $t = 2, \ldots, T$. The exact counterparts are case DU with $n = T-1$ (at $\phi = 1$ the distribution with an intercept does not depend on y_0) and case DK with $n = T$ and $z_0 = 0$. The latter holds because y_1 is itself a standard normal innovation, so the regression over $t = 2, \ldots, T$ coincides with the regression over

$t = 1, \ldots, T$ of a random walk started at $y_0 = 0$; in this case the statistic is $(T-1)/T$ times $n(\hat{\phi} - 1)$.

Three conclusions emerge. First, with an intercept the tabulated critical values agree with the exact ones to within 0.05, the simulation and interpolation error one expects from five million replications (the standard error of a simulated 5% quantile at $T = 100$ is 0.0074). The exact size of the nominal 5% test is between 4.98 and 5.03%. Second, without an intercept the tabulated values are systematically too negative at small T. At $T = 25$ the 5% critical value is -7.332 against the exact -7.076. The exact size of the nominal 5% test is 4.56% at $T = 25$, 4.77% at $T = 50$ and 4.88% at $T = 100$; 200,000 simulated random walks give 4.66%, 4.71% and 4.89%, within about two simulation standard errors, each about 0.05 percentage points. The tabulated values agree with the exact quantiles of a regression over T_e observations started at a known zero (case DK with $n = T_e$, $z_0 = 0$). In the data to which the test is applied, however, the first lagged value y_1 is random and contributes to the sums. The discrepancy is thus a one-observation shift, of the same kind as the one in Fuller's intercept table noted in Section 7.2. This is an inference, since the documentation does not say how the tables were generated; the exact sizes are facts about the stated experiment. Third, the tabulated p-values are truncated at 0.001, whereas the exact p-value of a statistic of -28.2 at $T = 500$ without intercept is 1.7×10^{-4}. A table lookup takes 0.9 ms and the inversion integral about 5 ms (Table 3), but a simulated 5% critical value accurate to ± 0.001 at $T = 100$ would need some 3×10^8 replications, about ten minutes, against under a tenth of a second for the exact quantile.

The exact moments also check earlier approximations and tables; Table S5 of the Online Supplement gives the bias, standard deviation, skewness and excess kurtosis of $\hat{\phi}$ in the four cases. For $\phi > 0$ and a fixed start, the bias in case DU is two to three times that in case DK, as the approximations $-(1 + 3\phi)/n$ and $-2\phi/n$ of Marriott and Pope (1954) (the first also in Kendall (1954)) predict. With a stationary start the ratio is similar except near the unit root: at $\phi = 0.99$ and $n = 25$ it is 5.4, because the stationary initial observation roughly halves the bias in case SK. Near the unit root, however, the approximations deteriorate in opposite directions: at $\phi = 0.99$ and $n = 25$ they give -0.159 and -0.079, where the exact biases are -0.194 (DU) and -0.065 (DK). Tables 3 and 4 of Vinod and Shenton (1996) give the mean, standard deviation, skewness and

kurtosis (not excess kurtosis) of $\hat{\phi}$ in cases DK ($z_0 = 0$) and SK for n from 8 to 300 and ϕ from 0.1 to 1 (their sample size in case SK counts the initial observation and corresponds to our $n+1$). Of their 360 entries, 356 agree with ours to all digits shown and one more to one unit in the last digit. The other three are the standard deviation, skewness and kurtosis at $\phi = 1$ and $n = 300$: 0.01066, -2.269 and 11.35 in their table, against our exact values 0.01050, -2.226 and 10.98 at the same n; their own extrapolation formulas (their (3.7)–(3.9)) give 0.01050, -2.228 and 10.99.

7.4 Exact versus asymptotic distributions

As Section 6 shows, the normal approximation is poor near the unit root. Table S8 of the Online Supplement shows that the local-to-unity approximation does better, though it too converges slowly in the left tail. For $c \in \{-10, -2, 2\}$, the exact 5% quantiles of $n(\hat{\phi} - \phi)$ differ from their limits by up to 17% at $n = 50$, 4.5% at $n = 200$ and 1% at $n = 1000$; at $c = 5$ the differences are 41–44%, 10% and 2% (with intercept, -2.16 against -1.51 at $n = 50$). The stationary-start limits (45)–(46) differ substantially from the fixed-start ones when $|c|$ is small, as Elliott (1999) emphasized. Figure 1 plots the exact densities of $n(\hat{\phi} - 1)$ at $n = 25$, 100 and 1000 against the Dickey–Fuller limits, and Figure S2 in the Online Supplement the densities of $\hat{\phi}$ in the four cases at $n = 25$.

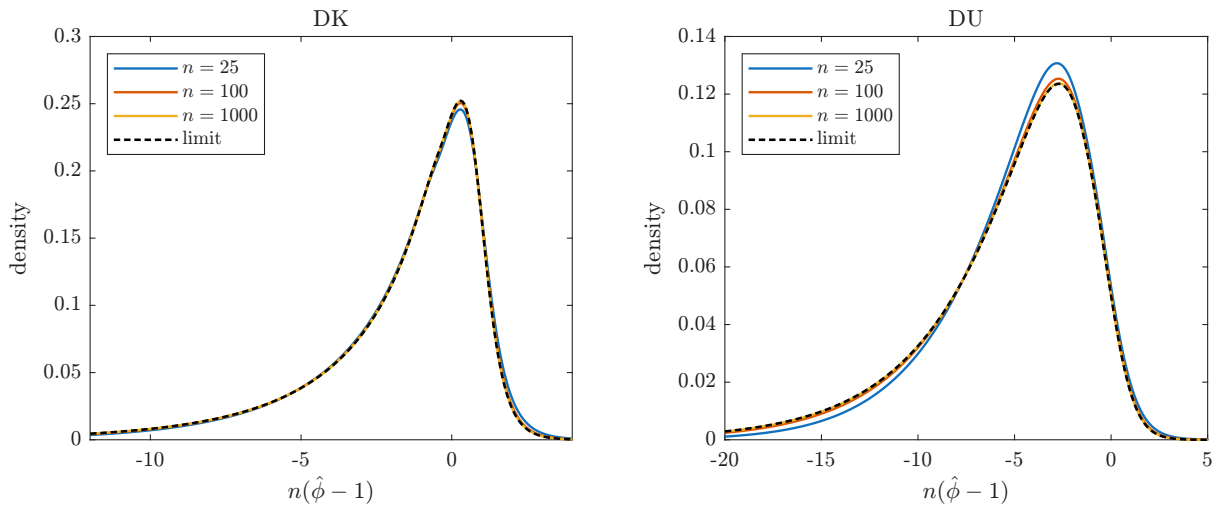


Figure 1: Exact densities of $n(\hat{\phi}-1)$ at $\phi = 1$ for $n = 25$, 100, 1000 and the Dickey–Fuller limits (dashed): DK with $z_0 = 0$ (left) and DU with $\gamma = 0$ (right).

Table S7 in the Online Supplement examines the random walk with drift (case DU, $\phi = 1$, $\gamma \neq 0$). The drift pulls the distribution of $n(\hat{\phi} - 1)$ toward zero and toward normality. At $n = 200$ and $\gamma = 1$, its standard deviation is 0.25 against 4.4 without drift, its mean -0.03 against -5.3, and its skewness -0.01. The local-to-unity limit evaluated at $\varkappa = \sqrt{n}\gamma$ describes the exact distribution well even for $\gamma = 0.1$, where neither the driftless Dickey–Fuller limit nor the normal limit of a dominant trend is adequate.

Figure 2 turns to the explosive region: it plots the exact densities of cases DK and DU at $\phi = 1.05$ against their common limit, the fast approach in case DK and the slow one in case DU of Section 6.2 side by side. Table S9 of the Online Supplement gives the exact quantiles of $|\phi|^n(\hat{\phi} - \phi)/(\phi^2 - 1)$ at $\phi = 1.05$ and 1.2, in case DK with $z_0 \in \{0, 1\}$ and in case DU with $\gamma = 0$, next to those of the limits. In case DK the convergence is governed by ϕ^n, not by n, and is far from monotone. At $\phi = 1.05$ and $z_0 = 0$, the exact 5% quantile is -10.4, -9.1, -13.6 and -16.4 at $n = 10$, 25, 50 and 100, before settling at -6.31, the Cauchy value, by $n = 400$. At $\phi = 1.2$ and $z_0 = 0$, the 5% and 95% quantiles at $n = 50$ are still 5–7% from the limit (the quartiles about 2%), and at $n = 100$ they agree with it to four digits. The row $n = 400$ at $\phi = 1.2$ ($\phi^{2n} \approx 2 \times 10^{63}$) can be computed only with the cancellation-free forms of Proposition 3: the determinant assembly of Proposition 2 and an eigenvalue evaluation both lose all accuracy there, as their terms are of order ϕ^{2n}. A fixed start $z_0 = 1$ changes the limit from Cauchy to the ratio of normals with the density f_∞ of Section 6.2, whose tails still decay like the inverse square of the argument but are lighter than the Cauchy tails by the factor $e^{-\lambda}$, $\lambda = z_0^2(\phi^2 - 1)/2$ (0.05 at $\phi = 1.05$ and 0.22 at $\phi = 1.2$); the exact distribution follows it closely from $n = 100$ on at $\phi = 1.2$. In case DU with $\gamma = 0$, whose limit is the same standard Cauchy distribution, the approach is the slow one of Section 6.2: at $\phi = 1.05$ the median is still -0.89 at $n = 100$ and -0.11 at $n = 400$, and at $\phi = 1.2$ it is -0.12 and -0.03, against the Cauchy value 0.

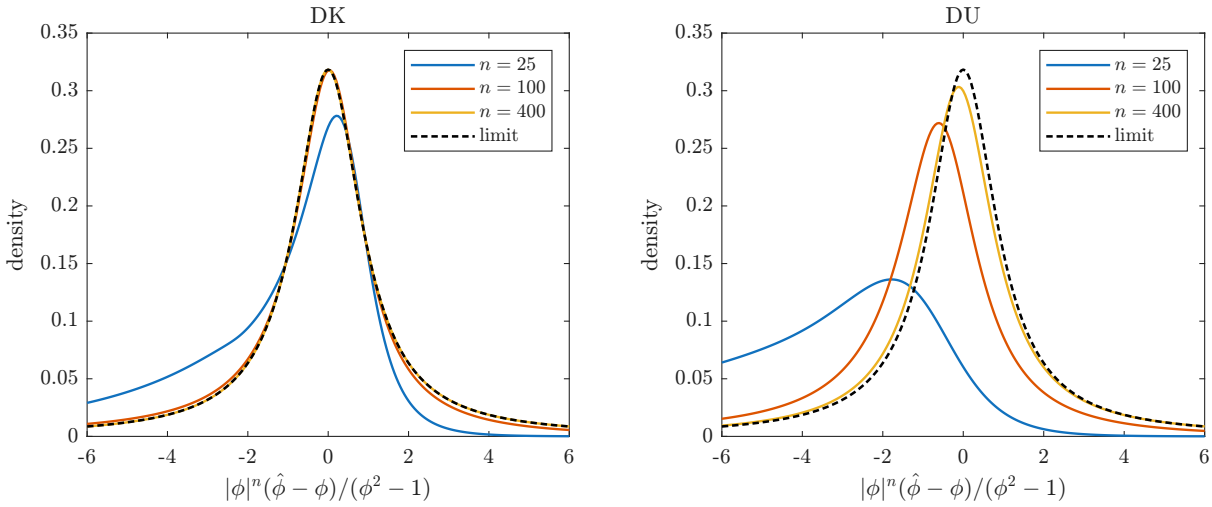


Figure 2: The explosive case $\phi = 1.05$: exact densities of $|\phi|^n(\hat{\phi} - \phi)/(\phi^2 - 1)$ for $n = 25$, 100, 400 and the standard Cauchy limit (dashed), in case DK with $z_0 = 0$ (left) and case DU with $\gamma = 0$ (right). The limit is reached like ϕ^{-n} in case DK and like $1/n$ in case DU.

8 Concluding Remarks

For all four standard cases of a Gaussian first-order autoregression, and for every real value of the coefficient with a fixed start, we show that the joint characteristic function of the numerator and denominator of the estimation error of the least squares estimator, and all the derivatives that the inversion and moment formulas require, are closed-form expressions in five scalars, which cost $O(1)$ operations per evaluation under the convention of Proposition 2. Chebyshev expansions, an accretivity theorem and rearranged explosive formulas address the principal numerical hazards. The resulting algorithms deliver exact distribution functions, densities, quantiles and moments, in finite samples and in the limits alike, together with numerical error diagnostics and documented finite-precision limits.

The method also extends to a regression with a linear time trend: detrending adds a rank-two correction to the Toeplitz core and two further primitives, and the cost remains $O(1)$. We carry out this analysis, including cancellation-free forms for the explosive region and the local-to-unity limits, in Section S7 of the Online Supplement; the MATLAB programs cover all the cases, including those with a trend.

Several uses suggest themselves. The exact median and quantile functions make the median-unbiased estimators and confidence intervals of Stock (1991) and Andrews (1993),

and the grid procedures of Hansen (1999) and Mikusheva (2007), computable without simulation at every sample size. These procedures are defined for a stationary start, or for a fixed start with $z_0 = 0$ (case DK) or $\gamma = 0$ (case DU); otherwise z_0 and γ are nuisance parameters that an exact procedure must fix, bound or profile out. The exact bias function is the binding function of the indirect-inference estimator of Phillips (2012). The exact distribution with a stationary versus a fixed initial condition quantifies the initial-condition sensitivity studied by Nabeya and Sørensen (1994), Müller and Elliott (2003), Hitomi et al. (2022) and Andrews et al. (2026). The MATLAB programs that implement every algorithm of the paper are released under the MIT license.[1] They come with a user manual, a validation suite, a driver that regenerates the computed entries of every table and figure in one command, and the 250-digit reference computation of Section 4.3.

Declaration of the Use of Generative AI

During the preparation of this work, the authors used Anthropic's Claude to assist with drafting and editing the exposition, with the implementation and testing of the accompanying MATLAB programs, and with checking derivations and numerical results. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

A Proofs

This appendix proves Lemmas 1 and 2, states and proves Lemma 3 on accretive matrices, and proves Theorem 2. Proposition 2 follows from the two lemmas and (19)–(20). The proof of Theorem 1, a direct computation of the Gaussian integrals, and those of Propositions 1 and 3 are in Section S6 of the Online Supplement, which also derives the limiting characteristic functions of Proposition 4 (Section S1).

[1] They are available at `https://raymond-kan.github.io/research.html`.

Proof of Lemma 1

With the two border coordinates permuted to the front, $\boldsymbol{\Sigma}$ is the Schur complement of $\boldsymbol{\Delta}$ in $\boldsymbol{K}$, so $|\boldsymbol{K}| = |\boldsymbol{\Delta}|\,|\boldsymbol{\Sigma}| = |\boldsymbol{\Delta}|\delta_4$, and by the block inverse formula $\boldsymbol{K}^{-1}$ has the blocks $\boldsymbol{\Sigma}^{-1}$ in the border coordinates, $-\boldsymbol{\Delta}^{-1}[\boldsymbol{a}\ \boldsymbol{b}]\boldsymbol{\Sigma}^{-1}$ off the diagonal, and $\boldsymbol{\Delta}^{-1}+\boldsymbol{\Delta}^{-1}[\boldsymbol{a}\ \boldsymbol{b}]\boldsymbol{\Sigma}^{-1}[\boldsymbol{a}\ \boldsymbol{b}]'\boldsymbol{\Delta}^{-1}$ in the core. With $\boldsymbol{g} = [\boldsymbol{a}\ \boldsymbol{b}]'\boldsymbol{\Delta}^{-1}\boldsymbol{\beta}_c$ and $\boldsymbol{h} = (0, \beta_n)'$ the border part of $\boldsymbol{\beta}$,

$$\begin{aligned}\boldsymbol{\beta}'\boldsymbol{K}^{-1}\boldsymbol{\beta} &= \boldsymbol{h}'\boldsymbol{\Sigma}^{-1}\boldsymbol{h} - 2\boldsymbol{h}'\boldsymbol{\Sigma}^{-1}\boldsymbol{g} + \boldsymbol{\beta}_c'\boldsymbol{\Delta}^{-1}\boldsymbol{\beta}_c + \boldsymbol{g}'\boldsymbol{\Sigma}^{-1}\boldsymbol{g} \\ &= \boldsymbol{\beta}_c'\boldsymbol{\Delta}^{-1}\boldsymbol{\beta}_c + (\boldsymbol{h}-\boldsymbol{g})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{h}-\boldsymbol{g}),\end{aligned}$$

and

$$\boldsymbol{\Sigma}^{-1} = \frac{1}{\delta_4}\begin{bmatrix}\delta_3 & -\delta_2 \\ -\delta_2 & \delta_1\end{bmatrix}$$

gives (16). □

Proof of Lemma 2

If $q = 0$, $\boldsymbol{D} = p\boldsymbol{I}_m$ and the statement is immediate. Otherwise, let $|\boldsymbol{D}_k|$ be the leading principal minor of order k; expanding $|\boldsymbol{D}_{k+1}|$ along its last row gives $|\boldsymbol{D}_{k+1}| = p|\boldsymbol{D}_k| - q^2|\boldsymbol{D}_{k-1}|$ with $|\boldsymbol{D}_0| = 1$, $|\boldsymbol{D}_1| = p$. When $p^2 \neq 4q^2$ the roots $r_\pm$ of $r^2 - pr + q^2$ are distinct and the solution of the recursion is $|\boldsymbol{D}_k| = (r_+^{k+1} - r_-^{k+1})/(r_+ - r_-)$. (Dividing the recursion by q^{k+1} shows that, with $x = p/(2q)$, the scaled minors $|\boldsymbol{D}_k|/q^k$ satisfy the recursion $U_{k+1}(x) = 2xU_k(x) - U_{k-1}(x)$ and the initial values $U_0(x) = 1$, $U_1(x) = 2x$ of the Chebyshev polynomials, so $|\boldsymbol{D}_k| = q^k U_k(x)$, as used in Section 4.1.) The $(1,1)$ entry of $\boldsymbol{D}^{-1}$ is $|\boldsymbol{D}_{m-1}|/|\boldsymbol{D}_m|$ and the $(1,m)$ entry is $(-1)^{m+1}q^{m-1}/|\boldsymbol{D}_m|$, because the cofactor of the $(1,m)$ entry is the determinant of a triangular matrix with diagonal q. These give d_3 and d_4. The row-sum identity $\boldsymbol{D}\mathbf{1}_m = (p+2q)\mathbf{1}_m - q(\boldsymbol{e}_{1,m} + \boldsymbol{e}_{m,m})$, multiplied by $\boldsymbol{e}_{1,m}'\boldsymbol{D}^{-1}$ and by $\mathbf{1}_m'\boldsymbol{D}^{-1}$ and using persymmetry, gives $(p+2q)d_2 = 1 + q(d_3 + d_4)$ and $(p+2q)d_1 = m + 2qd_2$. Substituting the Chebyshev forms of d_3 and d_4 and $p + 2q = 2q(x+1)$ gives (27); its numerators vanish at $x = -1$ to first and second order because $U_k(-1) = (-1)^k(k+1)$ and $U_k'(-1) = (-1)^{k+1}k(k+1)(k+2)/3$. □

A lemma on accretive matrices

The proofs of Theorem 2 and of the branch lemmas of the Online Supplement rest on three properties of accretive matrices, which we collect in one lemma.

Lemma 3 (Accretive matrices)**.** *Let $\boldsymbol{A}$ be an accretive matrix, that is, complex symmetric with $\boldsymbol{R} = \operatorname{Re}\boldsymbol{A} = (\boldsymbol{A} + \boldsymbol{A}^*)/2$ positive definite.*

(a) Every eigenvalue of $\boldsymbol{A}$ has positive real part; in particular $\boldsymbol{A}$ is nonsingular.

(b) $\boldsymbol{A}^{-1}$ is accretive. Hence $\operatorname{Re}(\boldsymbol{y}'\boldsymbol{A}^{-1}\boldsymbol{y}) > 0$ for every nonzero real vector $\boldsymbol{y}$, and for real b the number $1 + \mathrm{i}b\,\boldsymbol{y}'\boldsymbol{A}^{-1}\boldsymbol{y}$ is never a nonpositive real number.

(c) If $\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{21}' \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{bmatrix}$, the Schur complement $\boldsymbol{S} = \boldsymbol{A}_{11} - \boldsymbol{A}_{21}'\boldsymbol{A}_{22}^{-1}\boldsymbol{A}_{21}$ is accretive, and when its order is one or two its determinant is not on the closed negative real axis.

Proof. (a) For an eigenpair $\boldsymbol{A}\boldsymbol{z} = \lambda\boldsymbol{z}$ with $\boldsymbol{z} \neq \boldsymbol{0}$, $\boldsymbol{z}^*\boldsymbol{A}\boldsymbol{z} = \lambda\|\boldsymbol{z}\|^2$ has real part $\boldsymbol{z}^*\boldsymbol{R}\boldsymbol{z} > 0$. (b) Since $\boldsymbol{A}$ is nonsingular and complex symmetric, so is $\boldsymbol{A}^{-1}$, and

$$\operatorname{Re}(\boldsymbol{A}^{-1}) = \tfrac{1}{2}\big(\boldsymbol{A}^{-1} + \boldsymbol{A}^{-*}\big) = \boldsymbol{A}^{-*}\,\tfrac{1}{2}(\boldsymbol{A}^* + \boldsymbol{A})\,\boldsymbol{A}^{-1} = \boldsymbol{A}^{-*}\boldsymbol{R}\boldsymbol{A}^{-1},$$

which is positive definite. For a real vector $\boldsymbol{y} \neq \boldsymbol{0}$, $\operatorname{Re}(\boldsymbol{y}'\boldsymbol{A}^{-1}\boldsymbol{y}) = \boldsymbol{y}'\operatorname{Re}(\boldsymbol{A}^{-1})\boldsymbol{y} > 0$, and $\operatorname{Im}(1 + \mathrm{i}b\,\boldsymbol{y}'\boldsymbol{A}^{-1}\boldsymbol{y}) = b\,\operatorname{Re}(\boldsymbol{y}'\boldsymbol{A}^{-1}\boldsymbol{y})$ vanishes only when $b = 0$, where the number equals 1. (c) $\boldsymbol{A}_{22}$ is accretive, hence nonsingular by (a). For $\boldsymbol{z} \neq \boldsymbol{0}$ of the order of $\boldsymbol{S}$ put $\boldsymbol{w} = -\boldsymbol{A}_{22}^{-1}\boldsymbol{A}_{21}\boldsymbol{z}$ and $\boldsymbol{x} = (\boldsymbol{z}', \boldsymbol{w}')'$. Since $\boldsymbol{A}_{21}\boldsymbol{z} + \boldsymbol{A}_{22}\boldsymbol{w} = \boldsymbol{0}$, the terms $\boldsymbol{w}^*\boldsymbol{A}_{21}\boldsymbol{z}$ and $\boldsymbol{w}^*\boldsymbol{A}_{22}\boldsymbol{w}$ of the block expansion of $\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}$ cancel, and

$$\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x} = \boldsymbol{z}^*\boldsymbol{A}_{11}\boldsymbol{z} + \boldsymbol{z}^*\boldsymbol{A}_{21}'\boldsymbol{w} = \boldsymbol{z}^*\boldsymbol{S}\boldsymbol{z},$$

whose real part $\boldsymbol{x}^*\boldsymbol{R}\boldsymbol{x}$ is positive. Thus $\boldsymbol{S}$ is accretive, its eigenvalues have arguments in $(-\pi/2, \pi/2)$ by (a), and when there are at most two of them the argument of their product lies in $(-\pi, \pi)$. (The last conclusion fails for larger orders, which is why the two border coordinates are eliminated together.) □

Proof of Theorem 2

(i) $\boldsymbol{K}$ is accretive by Theorem 1, and so is $\boldsymbol{D}$, because Re $\boldsymbol{D}$ is a principal submatrix of Re $\boldsymbol{K}$ (in cases DU and SU the core block of Re $\boldsymbol{K}$ is Re $\boldsymbol{\Delta}$ = Re $\boldsymbol{D}$, since κ is purely imaginary). Hence $|\boldsymbol{K}| \neq 0$ and $|\boldsymbol{D}| \neq 0$ by Lemma 3(a). The eigenvalues of $\boldsymbol{D}$ are $\lambda_j = p + 2q\cos(j\pi/n)$, $j = 1, \ldots, m$, with

$$\text{Re } \lambda_j = 1 + \phi^2 - 2\phi\cos(j\pi/n) \geq (1 - |\phi|)^2 + 2|\phi|\big(1 - |\cos(j\pi/n)|\big) > 0,$$

so each $\operatorname{Arg}\lambda_j \in (-\pi/2, \pi/2)$ is a continuous function of (t_1, t_2), $\Phi = \sum_{j=1}^m \operatorname{Arg}\lambda_j$ is continuous,

$$|\boldsymbol{D}| = \prod_{j=1}^m \lambda_j = |\det \boldsymbol{D}|e^{\mathrm{i}\Phi},$$

and at $t_1 = t_2 = 0$ all λ_j are positive so $\Phi(0,0) = 0$.

(ii) By (19), $\rho = 1 + \kappa d_1$ with $d_1 = \mathbf{1}_m'\boldsymbol{D}^{-1}\mathbf{1}_m$ and $\kappa = \mathrm{i}b$, $b = 2[(1-\phi)t_1 + t_2]/n$ real; since $\boldsymbol{D}$ is accretive, Lemma 3(b) shows that ρ is never a nonpositive real number. For δ_4: $\boldsymbol{\Delta} = \boldsymbol{D} + \kappa\mathbf{1}_m\mathbf{1}_m'$ has Re $\boldsymbol{\Delta}$ = Re $\boldsymbol{D}$, so it is accretive, and $\delta_4 = |\boldsymbol{\Sigma}|$ is the determinant of the 2×2 Schur complement of $\boldsymbol{\Delta}$ in the accretive matrix $\boldsymbol{K}$ (Lemma 1), which by Lemma 3(c) is not on the closed negative real axis.

(iii) Write $P_+ = p + 2q$ and $P_- = p - 2q$, so that $\text{Re}\, P_+ = (1-\phi)^2 \geq 0$ and $\text{Re}\, P_- = (1+\phi)^2 \geq 0$ by (26), with at least one of the two strictly positive, and $P_\pm \neq 0$ by assumption. The principal roots $s_+ = P_+^{1/2}$ and $s_- = P_-^{1/2}$ therefore have $\operatorname{Arg} s_+, \operatorname{Arg} s_- \in [-\pi/4, \pi/4]$, at least one of them in the open interval. With $r_\pm = (s_+ \pm s_-)^2/4$ and $h = s_+s_-$ as in (30),

$$r_+ + r_- = \frac{s_+^2 + s_-^2}{2} = p, \qquad r_+r_- = \frac{(s_+^2 - s_-^2)^2}{16} = q^2, \qquad r_+ - r_- = s_+s_- = h,$$

so $r_\pm$ are the two roots of $r^2 - pr + q^2 = 0$ and $h^2 = P_+P_- = p^2 - 4q^2$.

Ordering. We have $|\operatorname{Arg} s_+ - \operatorname{Arg} s_-| \leq \pi/2$, with equality only if $\operatorname{Arg} s_+$ and $\operatorname{Arg} s_-$ are $\pm\pi/4$ with opposite signs. That would require $\text{Re}\, P_+ = \text{Re}\, P_- = 0$, i.e. $\phi = 1$ and

$\phi = -1$ at once. Hence

$$|s_+ + s_-|^2 - |s_+ - s_-|^2 = 4\,\mathrm{Re}(s_+\bar{s}_-) = 4|s_+||s_-|\cos(\mathrm{Arg}\, s_+ - \mathrm{Arg}\, s_-) > 0,$$

so $|r_+| > |r_-|$ and $|\zeta| < 1$, and therefore $\mathrm{Re}(1-\zeta^n) > 0$.

Positions. $\mathrm{Arg}(s_+ + s_-) \in (-\pi/4, \pi/4)$, since $s_+ + s_-$ is the sum of two nonzero vectors of the closed convex sector $|\mathrm{Arg}\, z| \le \pi/4$ at least one of which lies strictly inside it, and such a sum lies strictly inside; hence $\mathrm{Arg}\, r_+ = 2\,\mathrm{Arg}(s_+ + s_-) \in (-\pi/2, \pi/2)$ and $\mathrm{Re}\, r_+ > 0$. Likewise $\mathrm{Arg}\, h = \mathrm{Arg}\, s_+ + \mathrm{Arg}\, s_- \in (-\pi/2, \pi/2)$, so h is the principal square root of $h^2 = p^2 - 4q^2$, which is therefore not on the closed negative real axis.

The identity (31). Let $G = \{(t_1, t_2) : P_+ \neq 0,\ P_- \neq 0\}$. On G the right side Ψ of (31) is continuous by the preceding paragraphs, and

$$e^{\mathrm{i}\Psi} = \frac{|\boldsymbol{D}|}{|\det \boldsymbol{D}|} = e^{\mathrm{i}\Phi}$$

by (18) written as $|\boldsymbol{D}| = r_+^n(1-\zeta^n)/h$, so $\Psi - \Phi \in 2\pi\mathbb{Z}$ is constant on each connected component of G. For $|\phi| \neq 1$, $G = \mathbb{R}^2$ and $\Psi(0,0) = 0 = \Phi(0,0)$ because p, q are real with $p > 2|q|$ there. For $|\phi| = 1$ and fixed (t_1, t_2) with $t_2 \neq 0$, both Ψ and Φ are continuous in ϕ in a neighborhood of ± 1 (the set $\{P_+ \neq 0, P_- \neq 0\}$ is open in (ϕ, t_1, t_2)) and agree for $|\phi| \neq 1$, so $\Psi = \Phi$ at $|\phi| = 1$ as well.

(iv) With $\boldsymbol{J}_m$ the $m \times m$ shift matrix, $\boldsymbol{D} = q(2x\boldsymbol{I}_m + \boldsymbol{J}_m + \boldsymbol{J}_m')$ and $\bar{\boldsymbol{D}} = q(-2\boldsymbol{I}_m + \boldsymbol{J}_m + \boldsymbol{J}_m')$ share the eigenvectors of $\boldsymbol{J}_m + \boldsymbol{J}_m'$, whose eigenvalues are $2\cos\theta_j$, $\theta_j = j\pi/n$, $j = 1, \dots, m$. Hence the eigenvalues of $\boldsymbol{\Xi} = \bar{\boldsymbol{D}}^{-1}\boldsymbol{D}$ are

$$\xi_j = \frac{x + \cos\theta_j}{\cos\theta_j - 1} = 1 - \frac{\delta_x}{1 - \cos\theta_j}, \qquad \delta_x = x + 1,$$

$|\boldsymbol{\Xi}| = \prod_{j=1}^m \xi_j = |\boldsymbol{D}|/|\bar{\boldsymbol{D}}| = U_m(x)/U_m(-1)$ by $|\boldsymbol{D}_k| = q^k U_k(x)$, and

$$\lambda_j = p + 2q\cos\theta_j = 2q(x + \cos\theta_j) = -2q\,(1 - \cos\theta_j)\,\xi_j.$$

Put

$$\varsigma_j = \frac{5}{2n^2(1 - \cos\theta_j)},$$

so that $|\xi_j - 1| < \varsigma_j$ on the band $|\delta_x| < 5/(2n^2)$. Since $1 - \cos\theta_j \geq 1 - \cos(\pi/n)$ and $n^2(1-\cos(\pi/n)) \geq \frac{9}{2}$ for $n \geq 3$, each $\varsigma_j \leq \frac{5}{9} < 1$, and the identity $\sum_{j=1}^m (1-\cos\theta_j)^{-1} = (n^2-1)/3$ gives

$$\sum_{j=1}^m \varsigma_j = \frac{5}{6}\,(1 - n^{-2}) < \frac{5}{6}.$$

A point of the open disk of radius $\varsigma < 1$ about 1 has positive real part and argument of modulus less than $\arcsin\varsigma$, the angle between the real axis and the tangent to the disk from the origin, and $\arcsin\varsigma \leq \pi\varsigma/2$ because arcsin is convex on $[0,1]$ with $\arcsin 0 = 0$ and $\arcsin 1 = \pi/2$. Hence $\operatorname{Re}\xi_j > 0$, $\operatorname{Arg}\xi_j \in (-\pi/2, \pi/2)$, and

$$\sum_{j=1}^m |\operatorname{Arg}\xi_j| < \frac{\pi}{2}\sum_{j=1}^m \varsigma_j < \frac{5\pi}{12} < \frac{\pi}{2},$$

so the principal argument of $|\boldsymbol{\Xi}|$ is $\sum_{j=1}^m \operatorname{Arg}\xi_j$. Next, $q \neq 0$, since $x = p/(2q)$ is defined on the band, and $-q$ is not a negative number: if $q = \varpi$ with $\varpi > 0$ then $\operatorname{Re}\lambda_j = -2\varpi(1-\cos\theta_j)\operatorname{Re}\xi_j < 0$, contradicting part (i). Since $\operatorname{Arg}\lambda_j \in (-\pi/2, \pi/2)$ by part (i), $\operatorname{Arg}\xi_j \in (-\pi/2, \pi/2)$, and

$$\operatorname{Arg}\lambda_j \equiv \operatorname{Arg}(-q) + \operatorname{Arg}\xi_j \pmod{2\pi}, \quad \operatorname{Arg}\lambda_j - \operatorname{Arg}\xi_j \in (-\pi,\pi), \quad \operatorname{Arg}(-q) \in (-\pi,\pi),$$

the congruence is an equality, and summing over $j = 1, \ldots, m$ gives (32). For the band around $x = 1$, conjugating by $\operatorname{diag}(-1, 1, -1, \ldots)$ turns $\boldsymbol{D}$ into the same matrix with q replaced by $-q$, which has the same determinant and eigenvalues and lies on the band around $x = -1$; the result there, with $-q$ in place of q, gives the statement for x near 1. □

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